SIFT Detector

Shree K. Nayar

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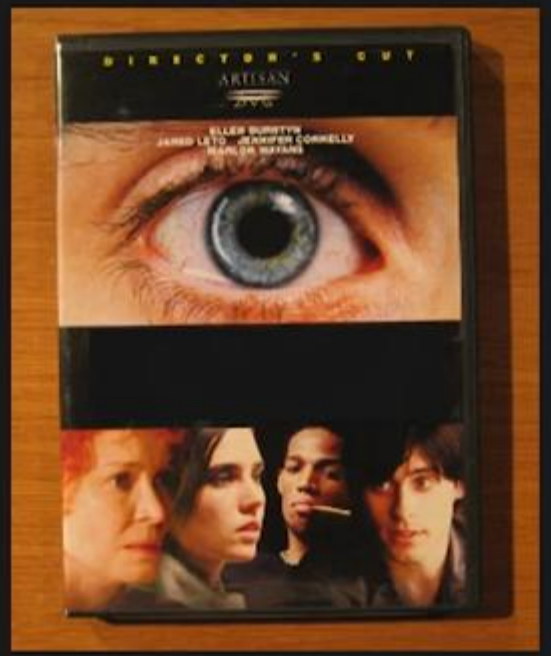
Topic: SIFT Detector, Module: Features

First Principles of Computer Vision

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A Little Quiz

How would you recognize the following types of objects?



Template



Rich 2D image

A Little Quiz

How would you recognize the following types of objects?



Template



Rich 2D image

Find and Match "Interesting Points or Features"

SIFT Detector

Scale Invariant Feature Transform (SIFT) and its use for image alignment and 2D object recognition.

Topics:

(1) What is an Interest Point?

(2) Detecting Blobs

(3) SIFT Detector

(4) SIFT Descriptor

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What is an Interest Point?

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Raw Images are Hard to Match





Different size, orientation, lighting, brightness, etc.

Removing Sources of Variation



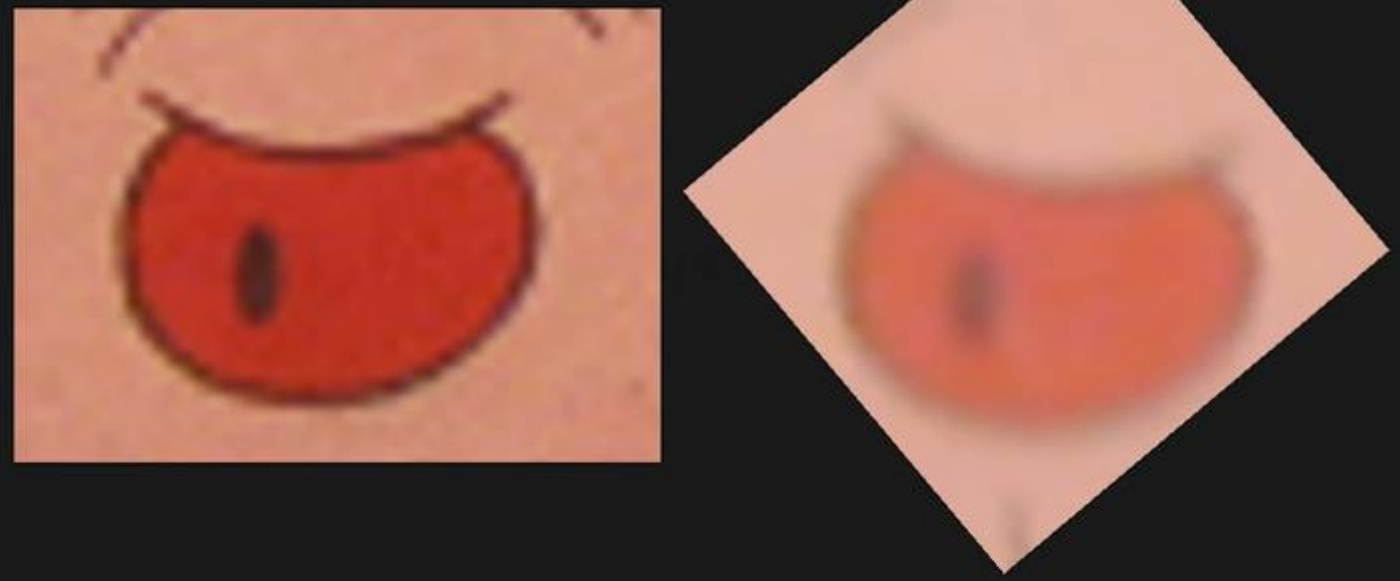


Removing Sources of Variation



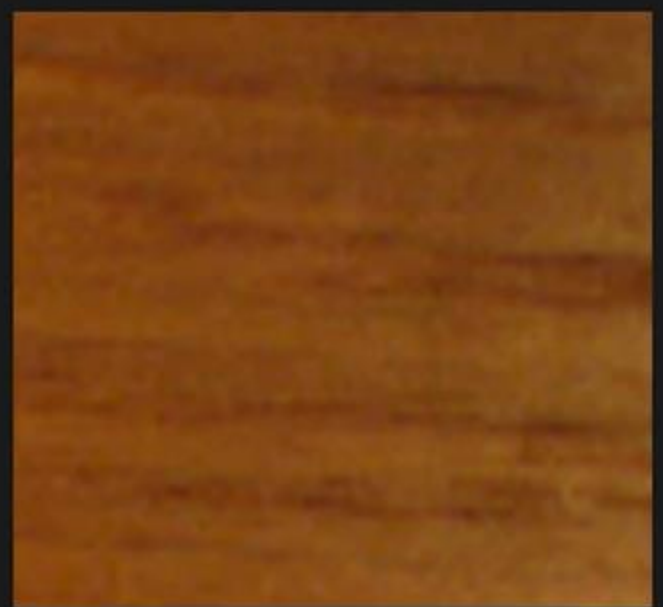


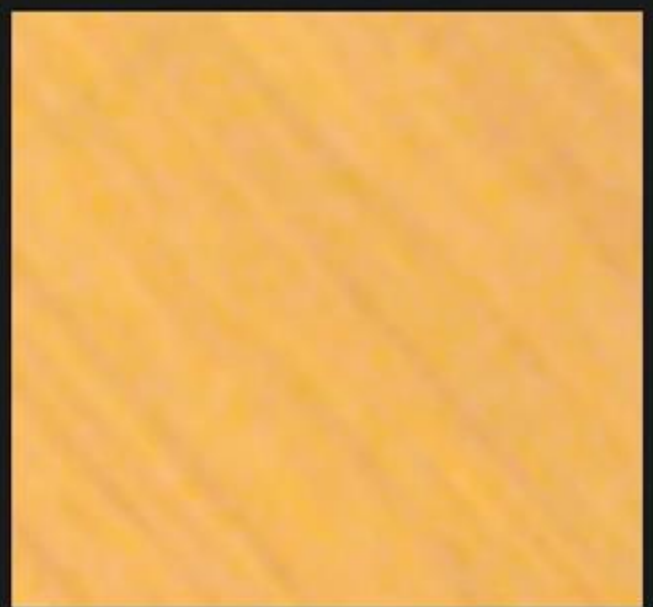
Removing Sources of Variation



Matching becomes easier if we can
remove variations like size and orientation.

Some Patches are not "Interesting"

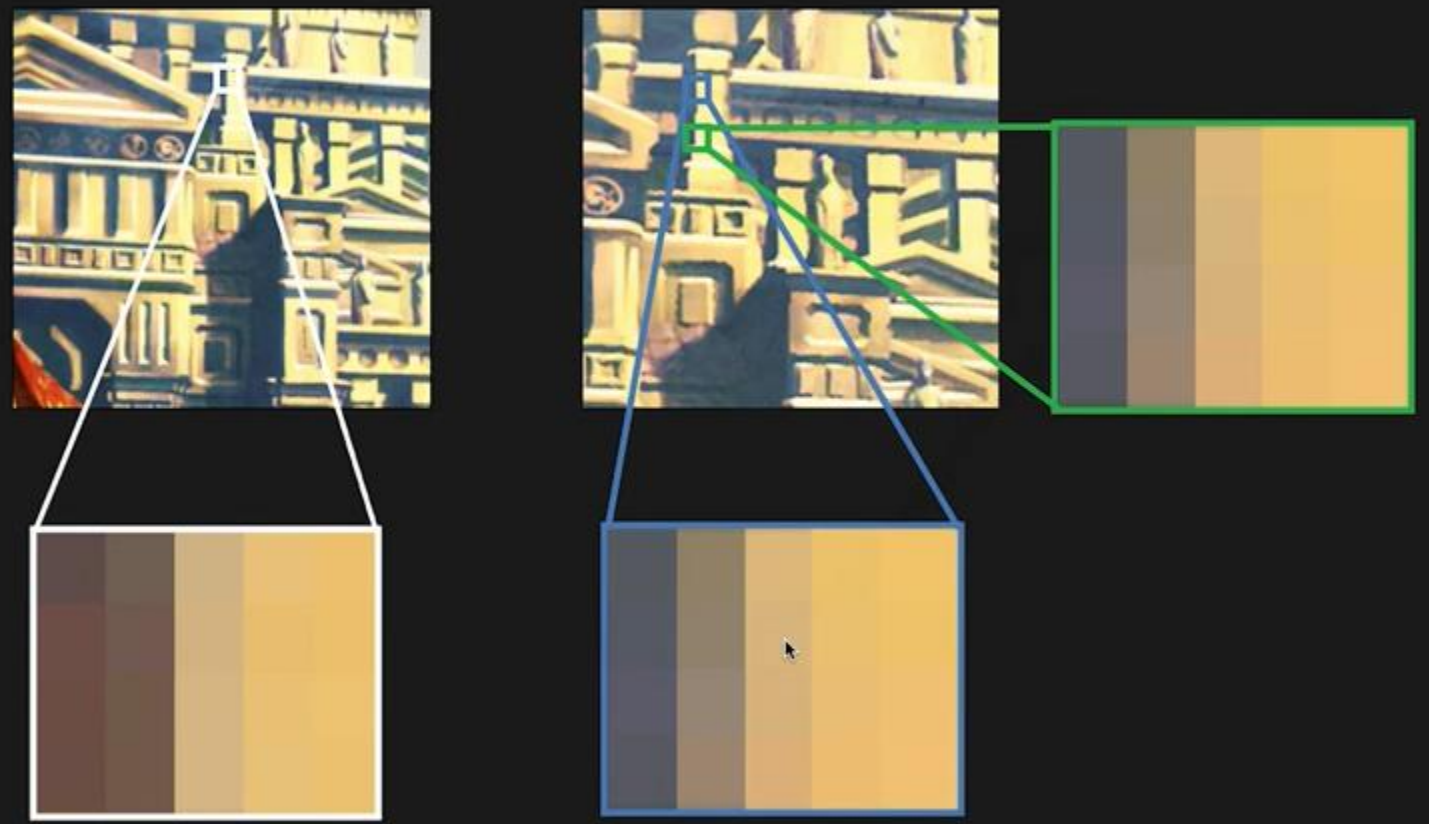




What is an Interesting Point/Feature?

- Has rich image content (brightness variation, color variation, etc.) within the local window
- Has well-defined representation (signature) for matching/comparing with other points
- Has a well-defined position in the image
- Should be invariant to image rotation and scaling
- Should be insensitive to lighting changes

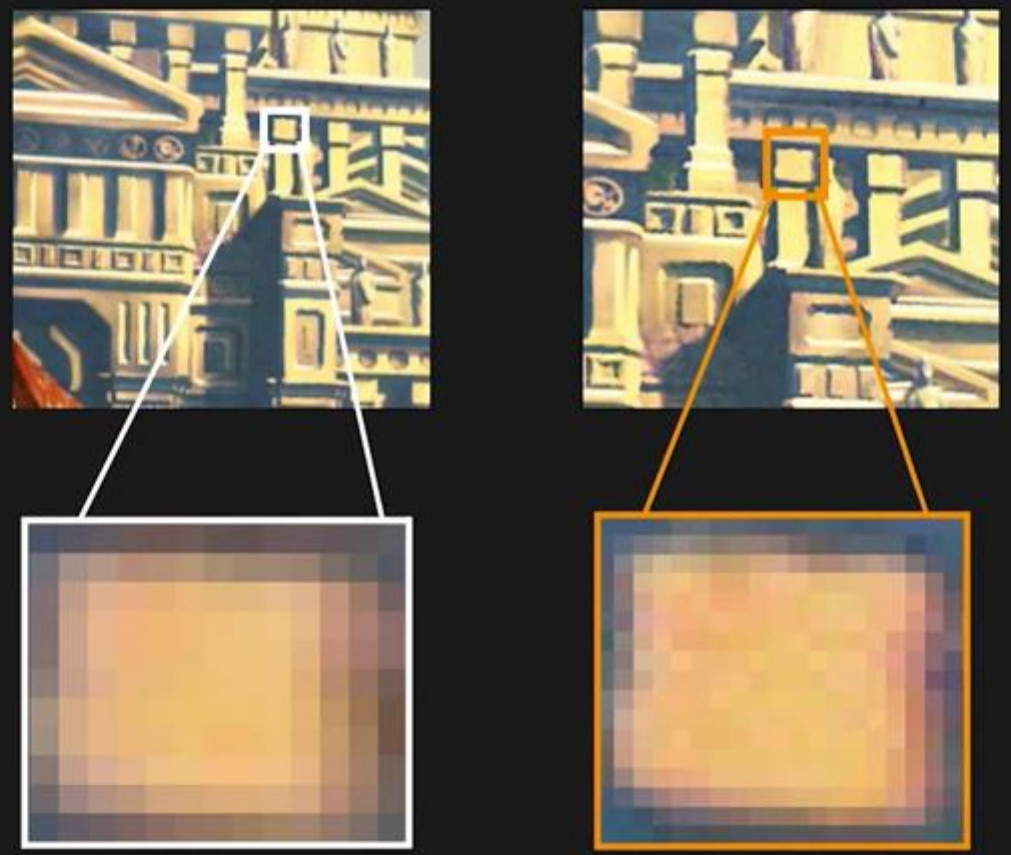
Are Lines/Edges Interesting?



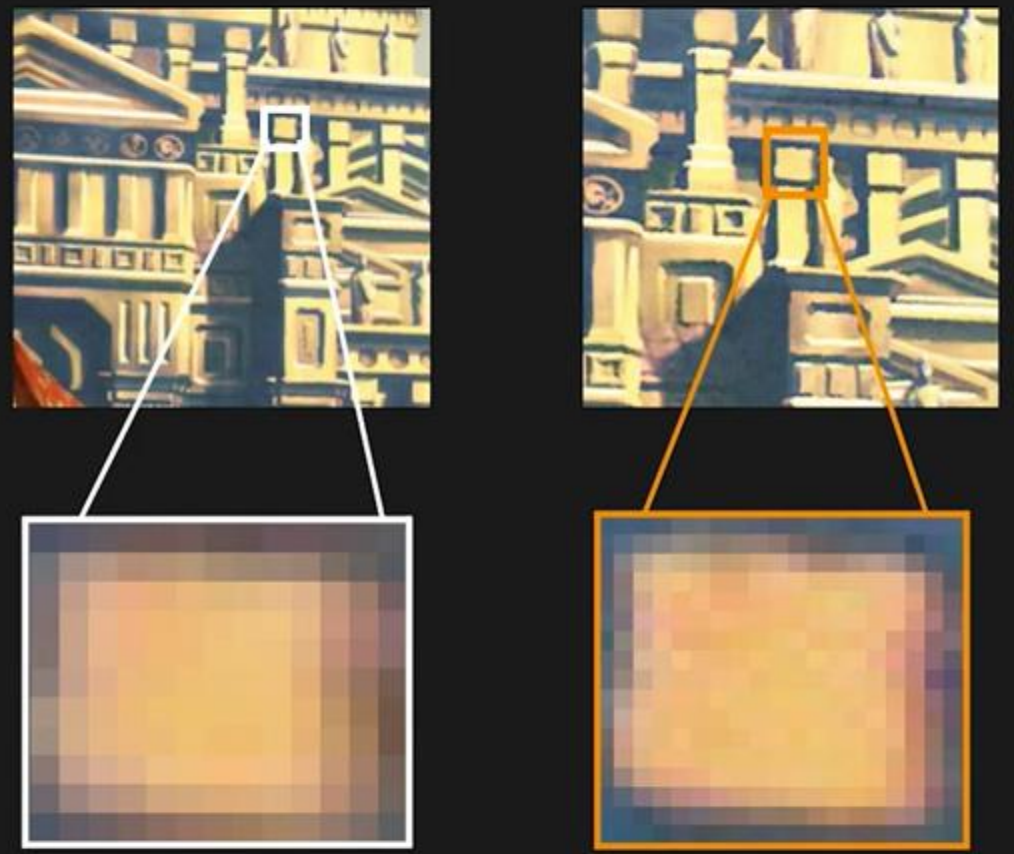
Cannot "Localize" an Edge

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Are Blobs Interesting?



Are Blobs Interesting?



Yes! Blobs have fixed position and definite size.

Blobs as Interest Points

For a Blob-like Feature to be useful, we need to:

- Locate the blob
- Determine its size
- Determine its orientation
- Formulate a description or signature that is independent of size and orientation



Detecting Blobs

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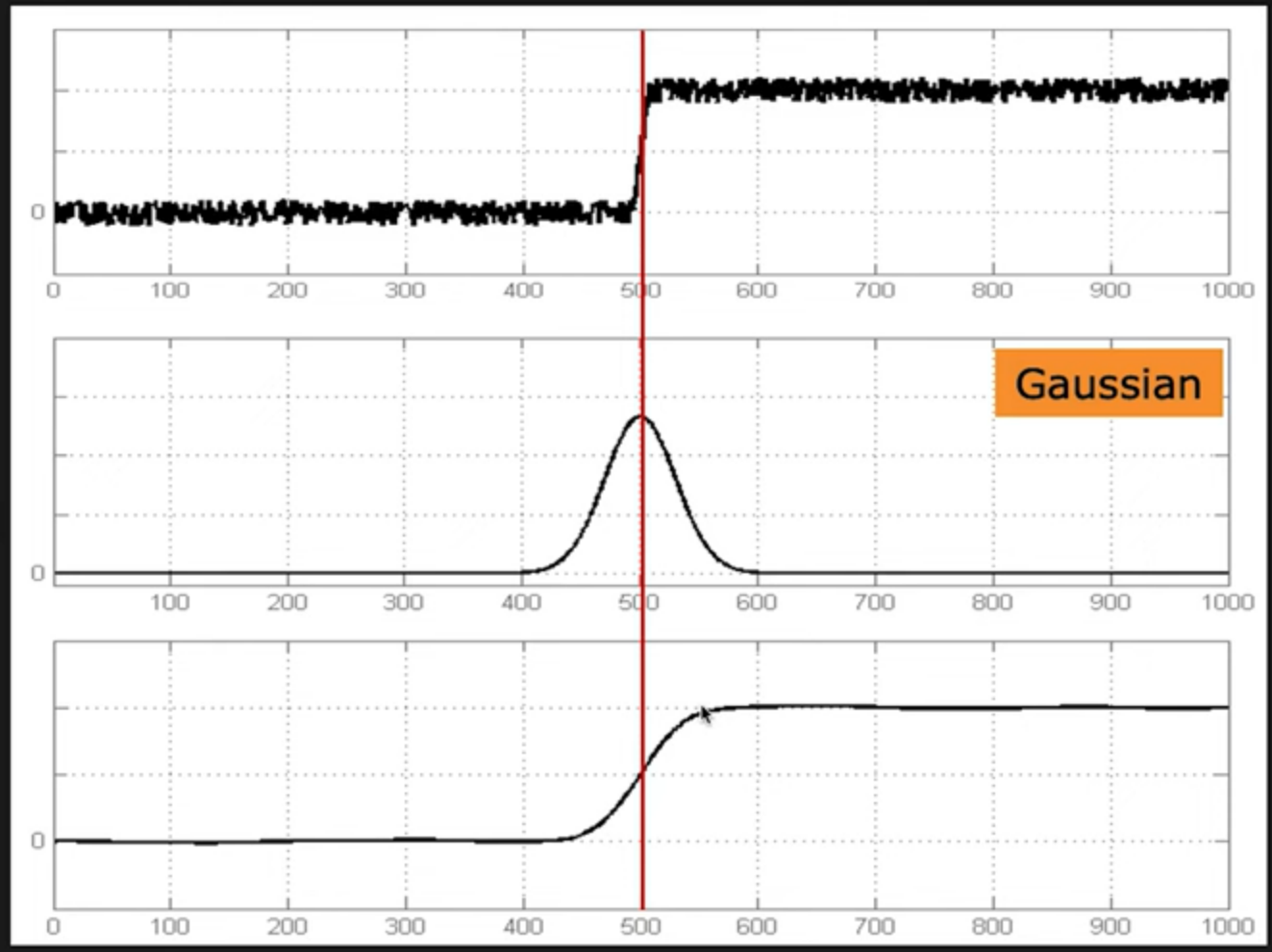
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Topic: SIFT Detector, Module: Features

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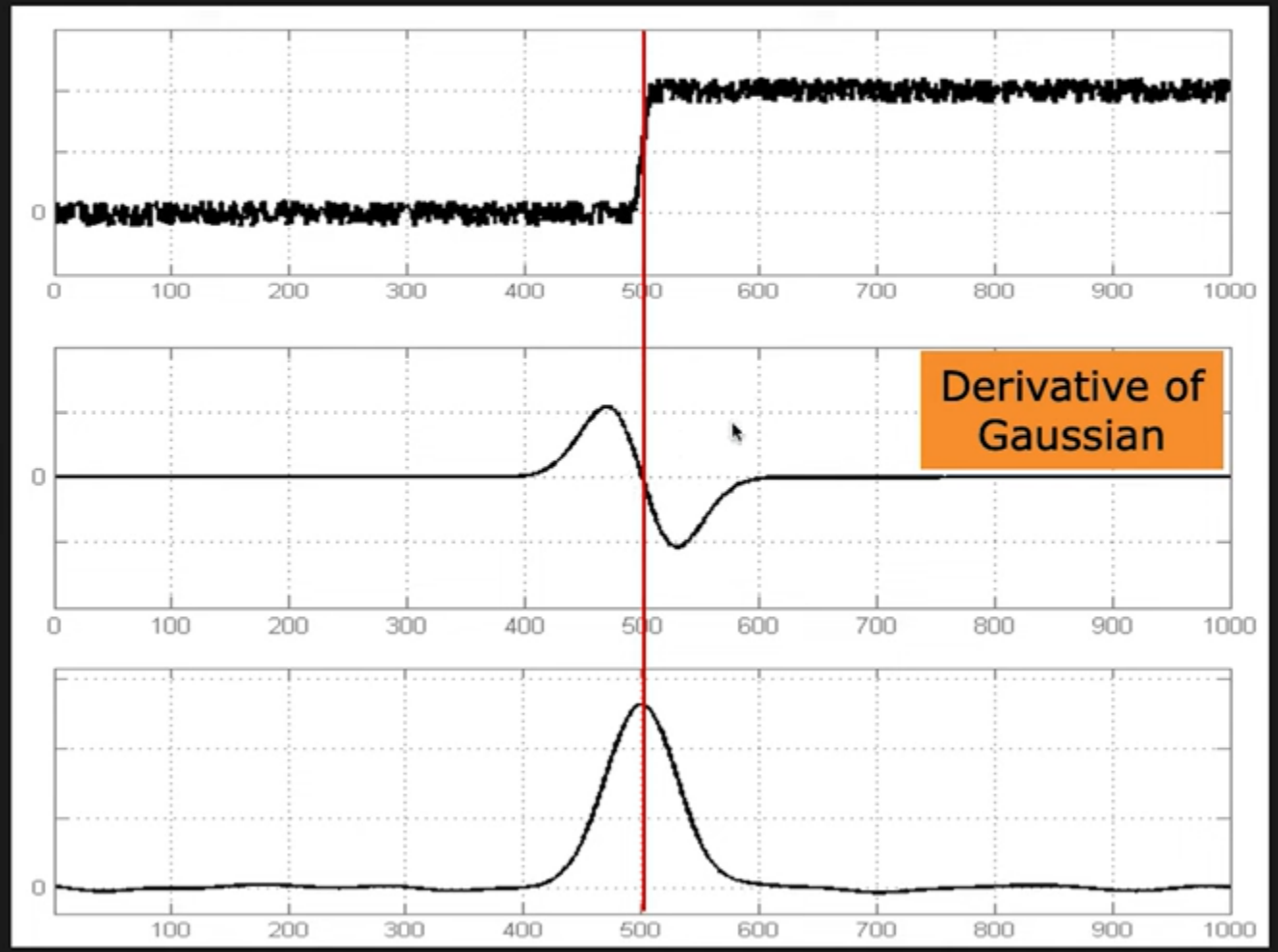
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Review: Gaussian Filter



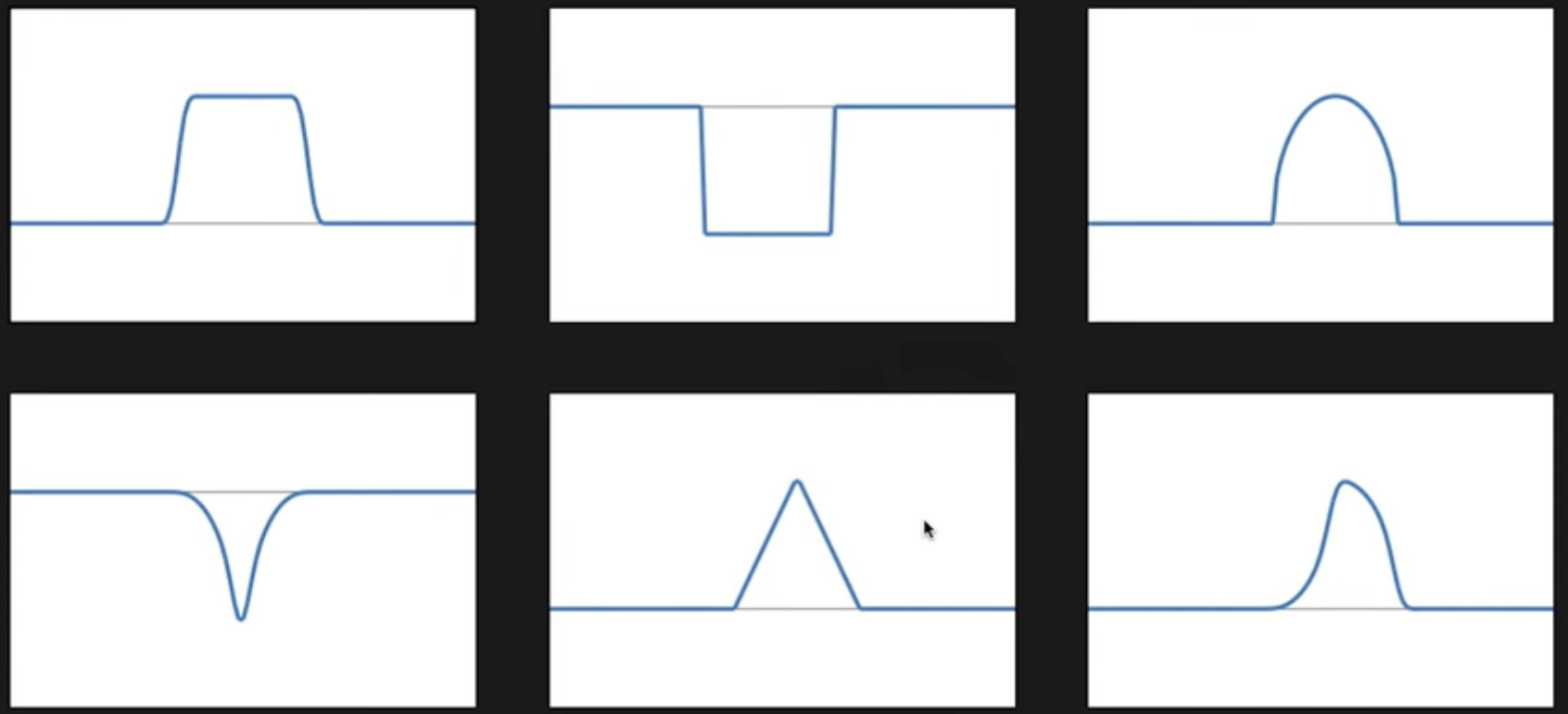
Gaussian Filter is used for removing noise by smoothing

Review: Derivative of Gaussian



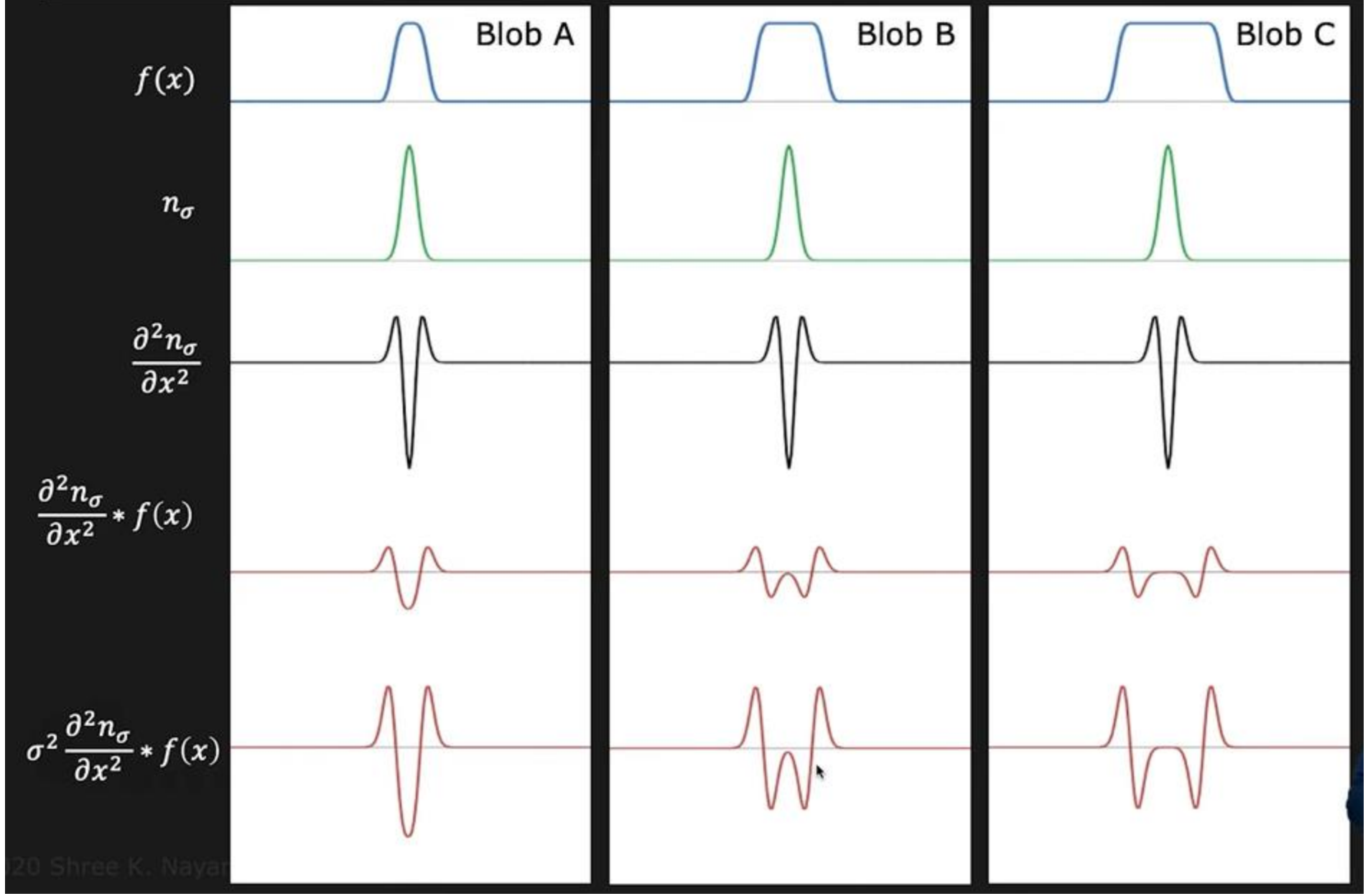
Extremum of Derivative of Gaussian denotes an Edge

1D Blobs



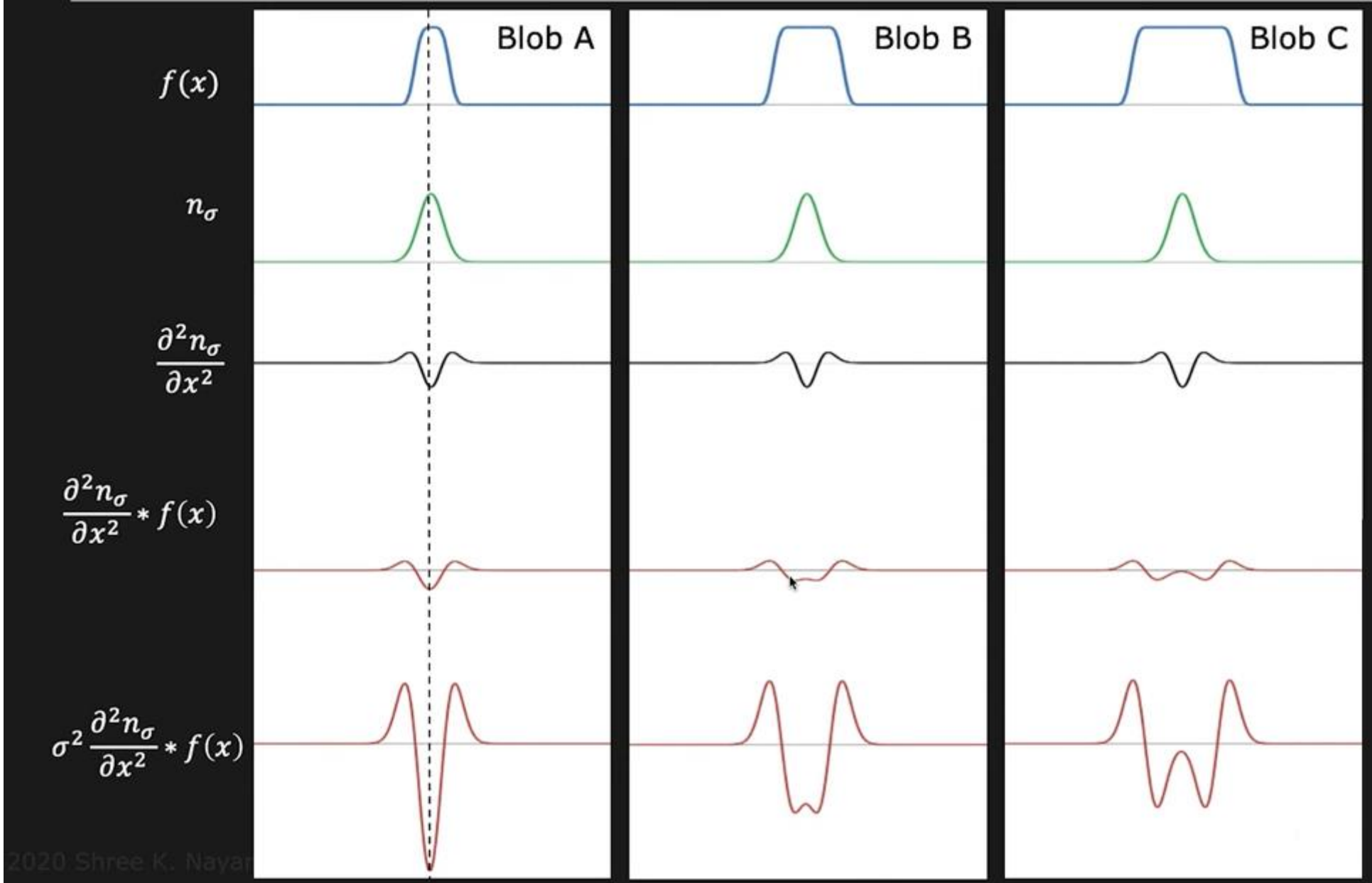
Examples of 1D Blob-like structures

1D Blob and 2nd Derivative of Gaussian



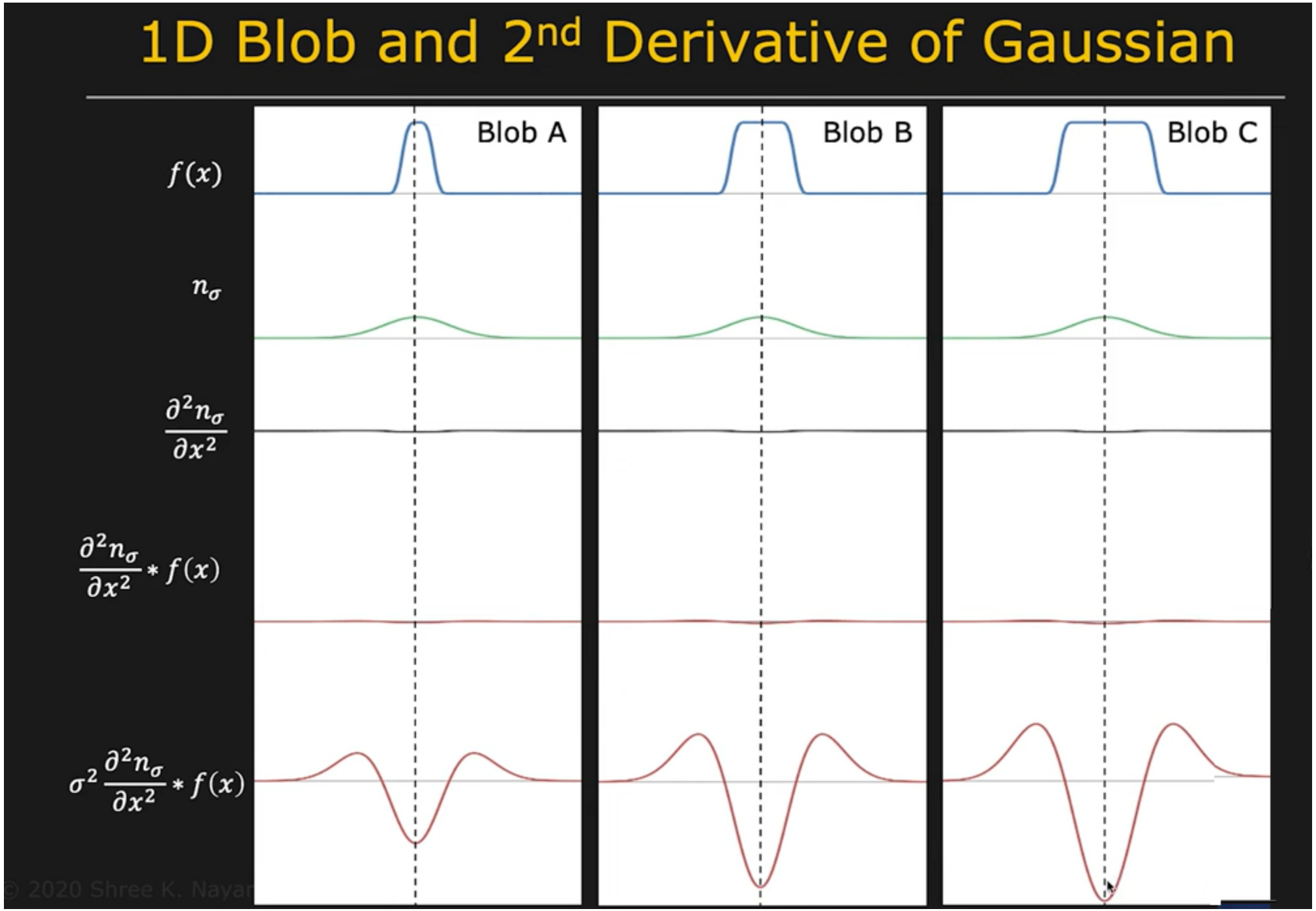
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1D Blob and 2nd Derivative of Gaussian



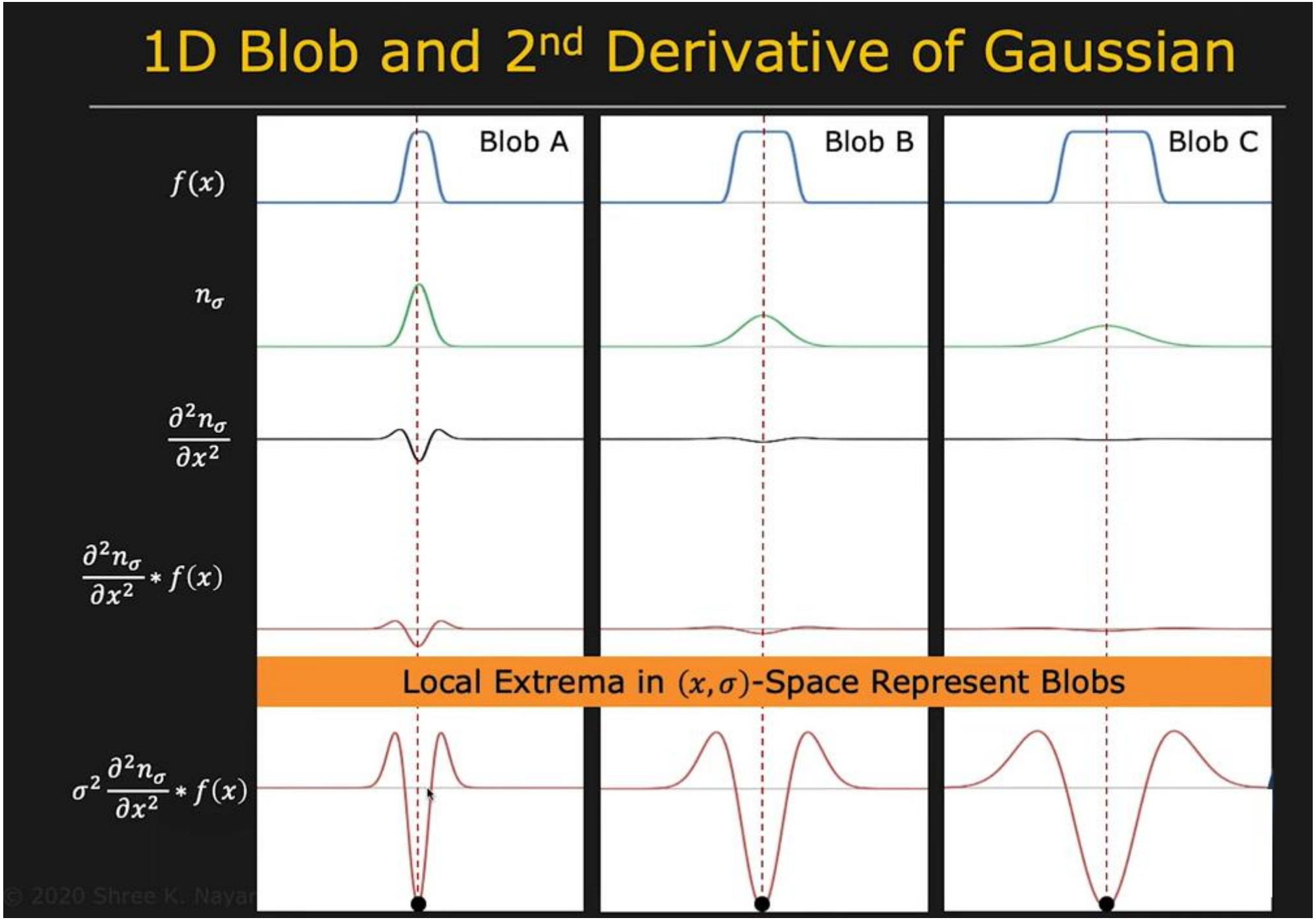
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1D Blob and 2nd Derivative of Gaussian



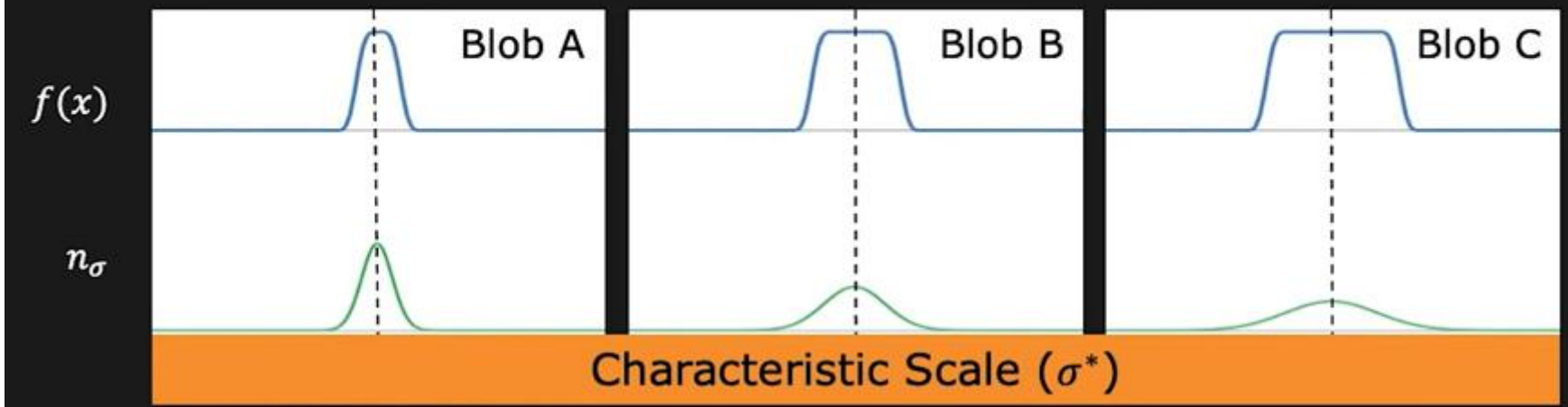
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1D Blob and 2nd Derivative of Gaussian



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Characteristic Scale and Blob Size



$\sigma_A{}^* = \sigma_1$ $\sigma_B{}^* = 2\sigma_1$ $\sigma_C{}^* = 3\sigma_1$

Characteristic Scale: The σ at which σ-normalized 2nd derivative attains its extreme value.

Characteristic Scale $\propto$ Size of Blob

$$\frac{\text{Size of Blob A}}{\text{Size of Blob B}} = \frac{\sigma_A{}^*}{\sigma_B{}^*}; \quad \frac{\text{Size of Blob B}}{\text{Size of Blob C}} = \frac{\sigma_B{}^*}{\sigma_C{}^*}$$

1D Blob Detection Summary

Given: 1D signal $f(x)$

Compute: $\sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x)$ at many scales $(\sigma_0, \sigma_1, \sigma_2, \ldots, \sigma_k)$.

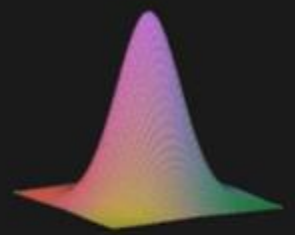
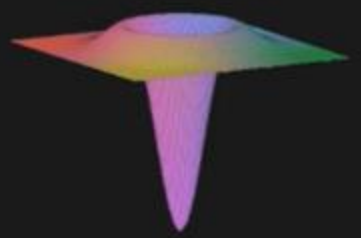
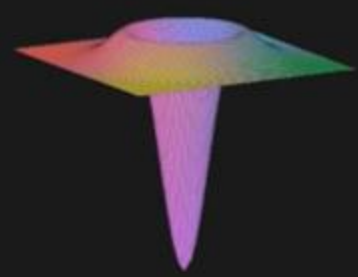
Find: $$(x^*, \sigma^*) = \arg\max_{(x,\sigma)} \left| \sigma^2 \frac{\partial^2 n_\sigma}{\partial x^2} * f(x) \right|$$

x^*: Blob Position

σ^*: Characteristic Scale (Blob Size)

2D Blob Detector

Normalized Laplacian of Gaussian (NLoG) is used as the 2D equivalent for Blob Detection.

Laplacian	Gaussian	LoG	NLoG
$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$			
	n_σ	$\nabla^2 n_\sigma$	$\sigma^2 \nabla^2 n_\sigma$

Location of Blobs given by Local Extrema after applying Normalized Laplacian of Gaussian at many scales.

Scale-Space






...

$S(x, y, \sigma_0)$ $S(x, y, \sigma_1)$ $S(x, y, \sigma_2)$ $S(x, y, \sigma_3)$

Increasing σ, Higher Scale, Lower Resolution

Scale Space: Stack created by filtering an image with Gaussians of different sigma (σ)

$$S(x, y, \sigma) = n(x, y, \sigma) * I(x, y)$$

Creating Scale-Space






...

$S(x, y, \sigma_0)$ $S(x, y, \sigma_1)$ $S(x, y, \sigma_2)$ $S(x, y, \sigma_3)$

Increasing σ, Higher Scale, Lower Resolution

Selecting sigmas to generate the scale-space:

$$\sigma_k = \sigma_0 s^k \qquad k = 0,1,2,3, \ldots$$

s: Constant multiplier

σ_0: Initial Scale

Blob Detection using Local Extrema



$S(x, y, \sigma_0)$ $S(x, y, \sigma_1)$ $S(x, y, \sigma_2)$ $S(x, y, \sigma_3)$ ...

$\sigma^2 \nabla^2 S(x, y, \sigma)$

$(NLoG * I(x, y))$



Blob Detection using Local Extrema






...

$S(x, y, \sigma_0)$ $S(x, y, \sigma_1)$ $S(x, y, \sigma_2)$ $S(x, y, \sigma_3)$

$\sigma^2 \nabla^2 S(x, y, \sigma)$

$(NLoG * I(x, y))$



2D Blob Detection Summary

Given an image $I(x, y)$

Convolve the image using NLoG at many scales σ

Find:

$$(x^*, y^*, \sigma^*) = \arg\max_{(x,y,\sigma)} |\sigma^2 \nabla^2 n_\sigma * I(x, y)|$$

(x^*, y^*): Position of the blob

σ^*: Size of the blob

SIFT Detector

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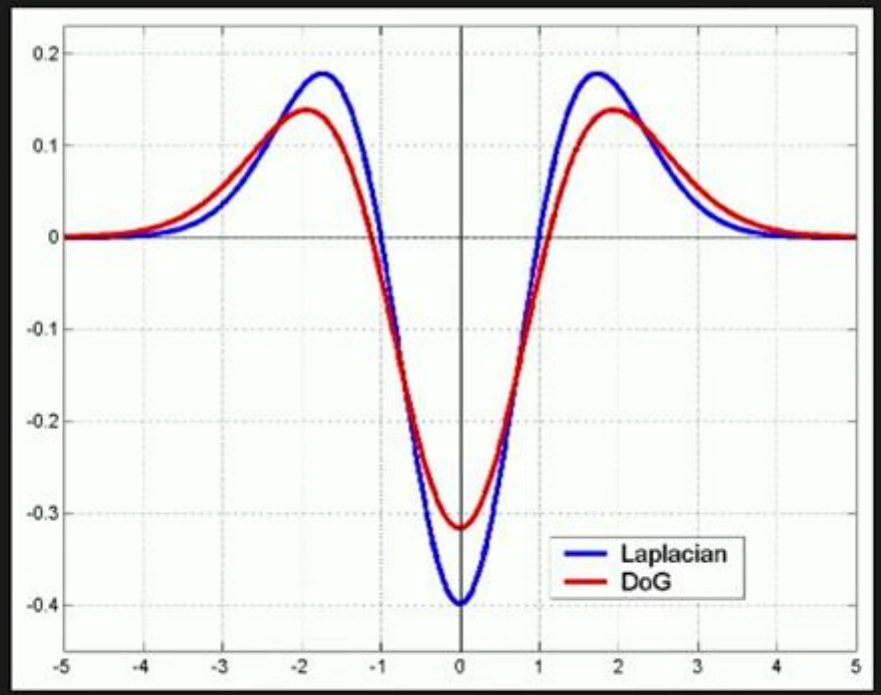
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Fast NLoG Approximation: DoG

Difference of Gaussian (DoG) $= (n_{s\sigma} - n_{\sigma}) \approx (s-1)\underbrace{\sigma^2 \nabla^2 n_{\sigma}}_{\text{NLoG}}$

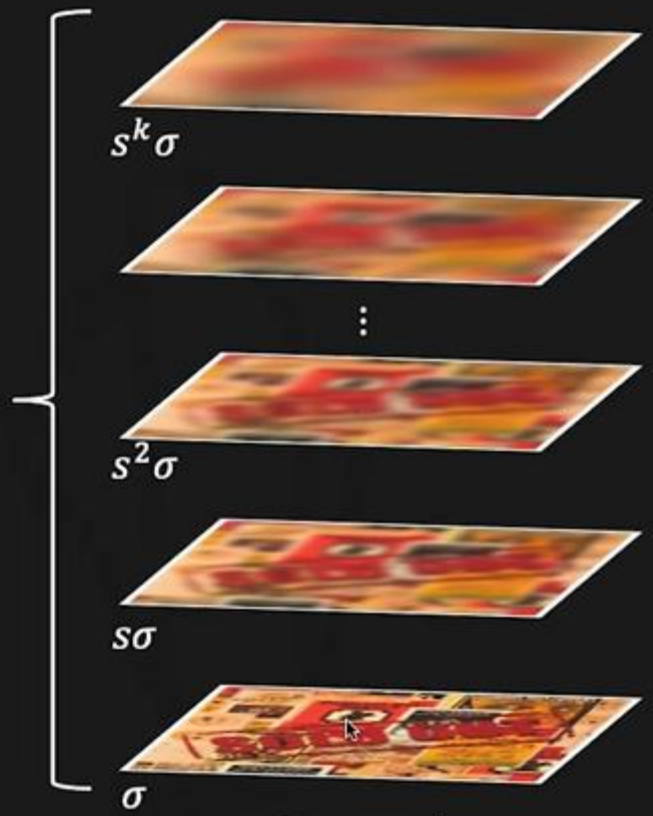


$\text{DoG} \approx (s-1)\ \text{NLoG}$

Extracting SIFT Interest Points



Image
$I(x, y)$

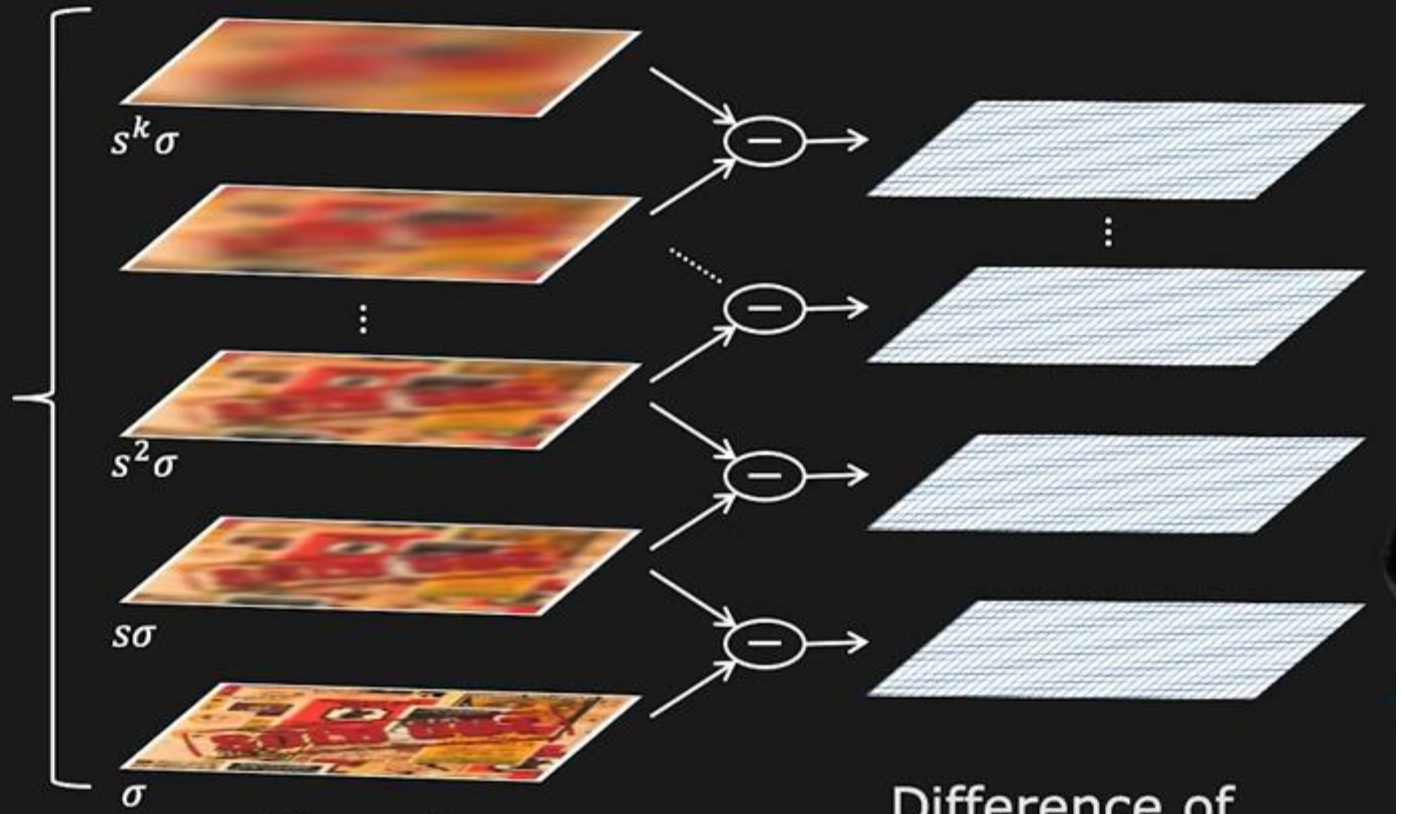


Gaussian
Scale-Space
$S(x, y, \sigma)$

Extracting SIFT Interest Points



Image
$I(x, y)$



Gaussian
Scale-Space
$S(x, y, \sigma)$

Difference of
Gaussians (DoG)
$\approx (s-1)\sigma^2\nabla^2 S(x, y, \sigma)$

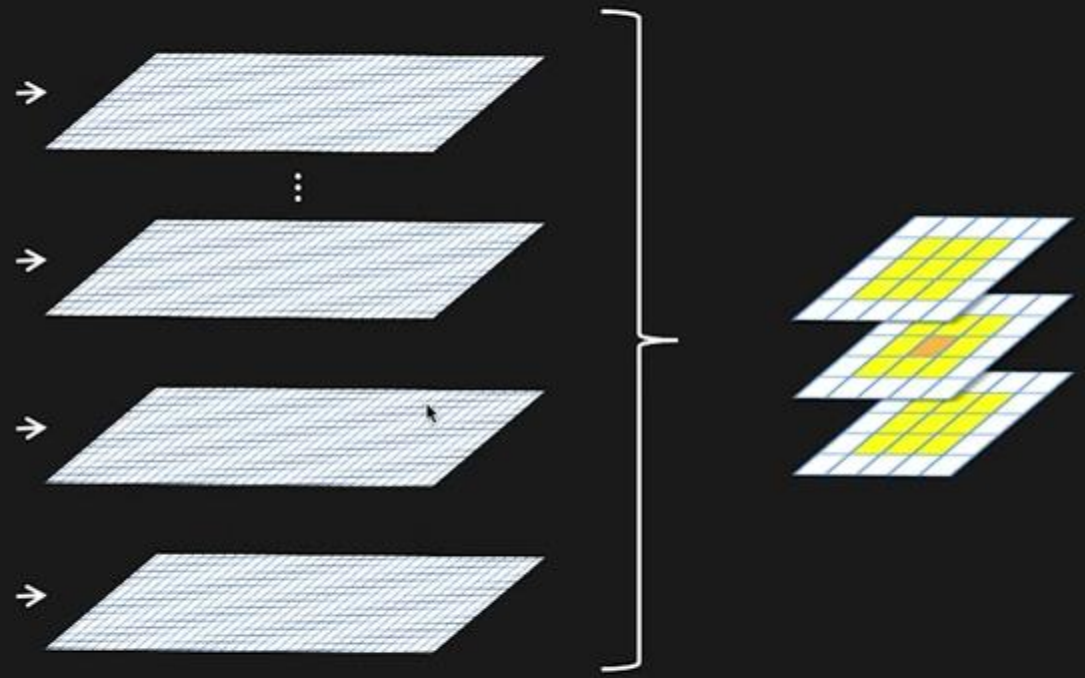
Extracting SIFT Interest Points



Difference of Gaussians (DoG)

$$\approx (s-1)\sigma^2\nabla^2 S(x,y,\sigma)$$

Extracting SIFT Interest Points

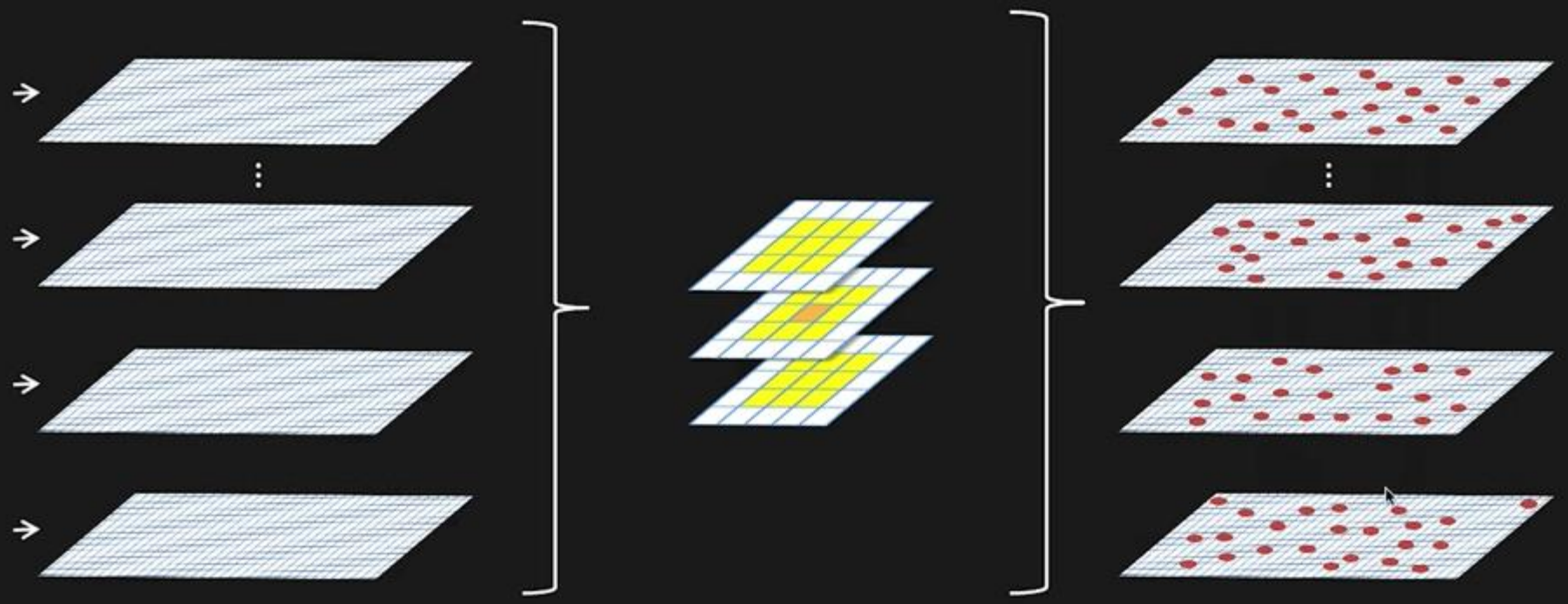


Difference of Gaussians (DoG)

$\approx (s-1)\sigma^2\nabla^2 S(x,y,\sigma)$

Find Extremum in every 3x3x3 grid

Extracting SIFT Interest Points



Difference of Gaussians (DoG)

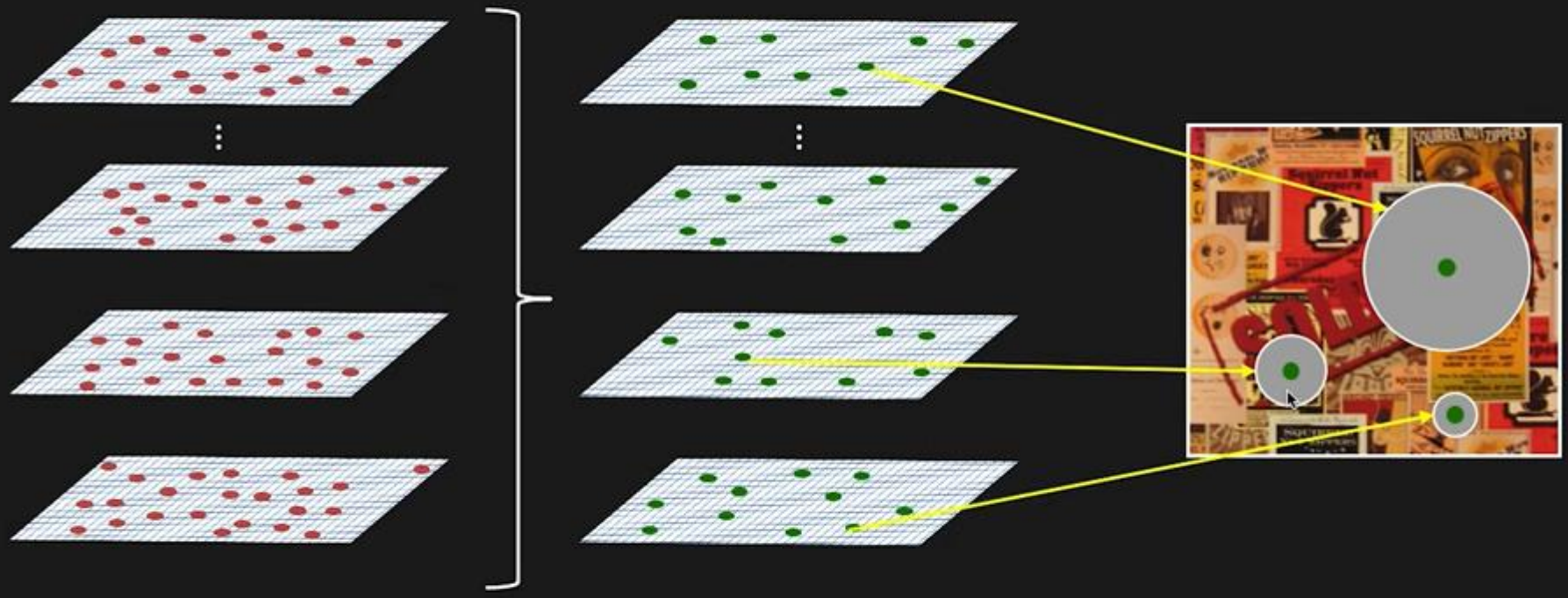
$\approx (s-1)\sigma^2\nabla^2 S(x,y,\sigma)$

Find Extremum in every 3x3x3 grid

Interest Point Candidates

(includes weak extrema)

Extracting SIFT Interest Points



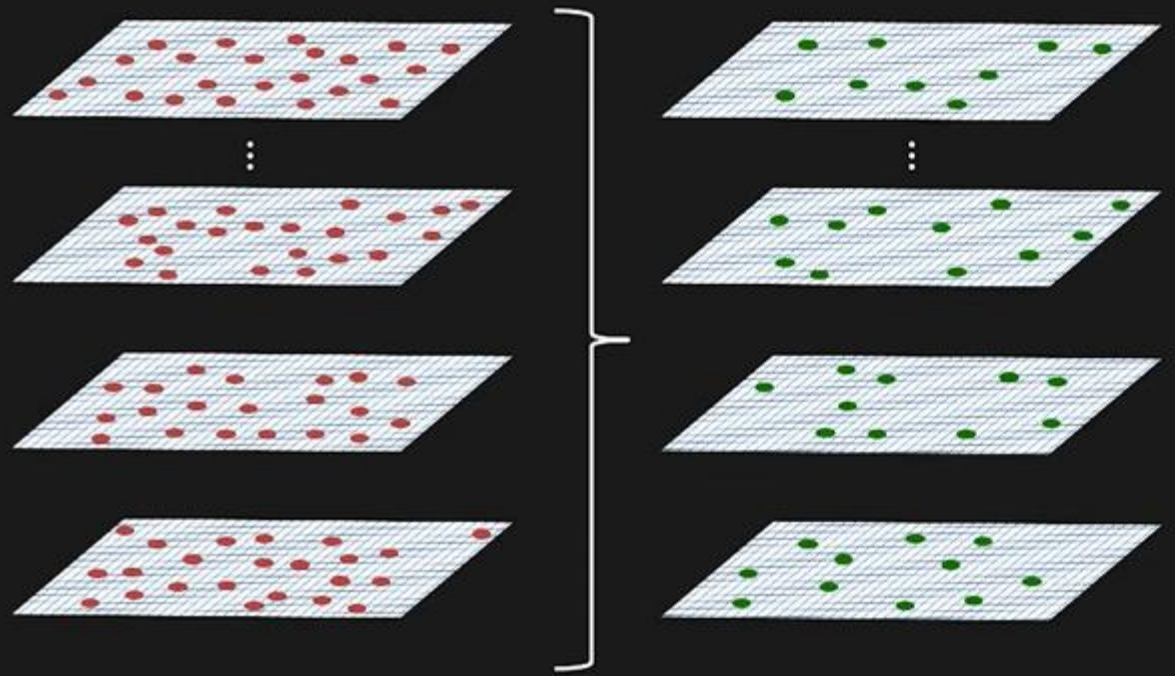
Interest Point Candidates

(includes weak extrema)

SIFT Interest Points

(after removing weak extrema)

Extracting SIFT Interest Points



Interest Point Candidates

(includes weak extrema)

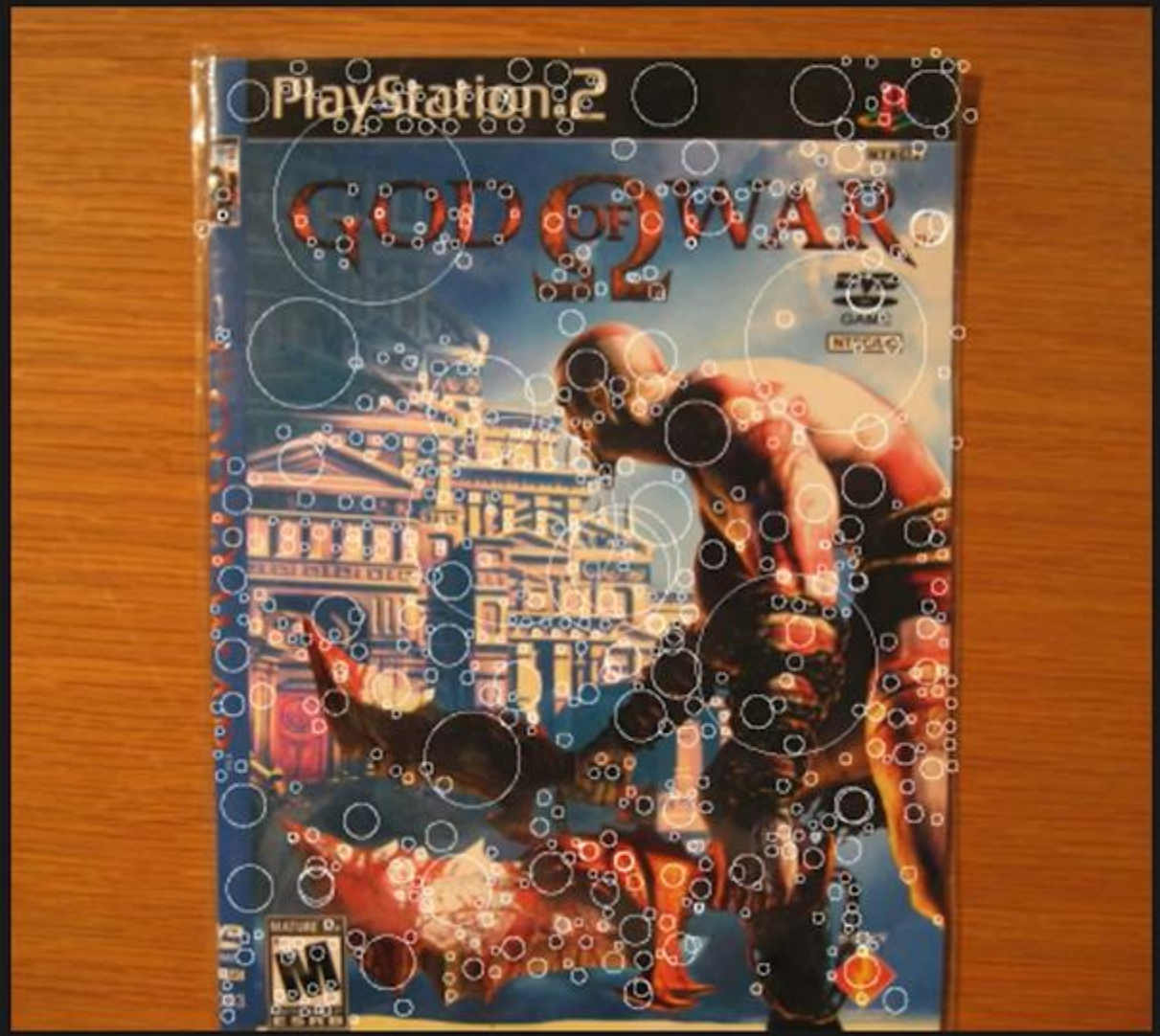
SIFT Interest Points

(after removing weak extrema)



Interest Point Visualization

SIFT Detection Examples



SIFT Detection Examples



SIFT Detection Examples



SIFT Scale Invariance



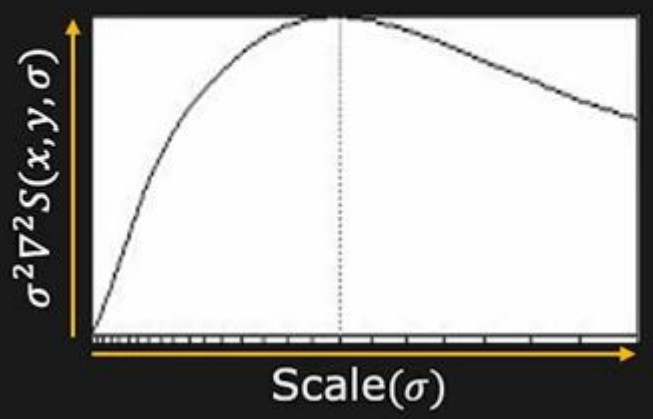


SIFT Scale Invariance







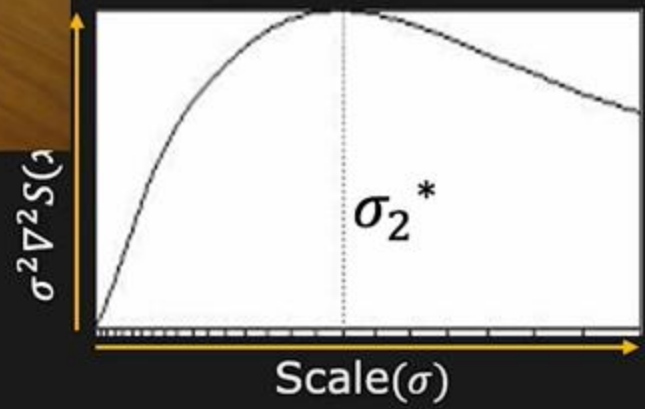


SIFT Scale Invariance









$\frac{\sigma_1{}^*}{\sigma_2{}^*}$: Ratio of Blob Sizes

Computing the Principal Orientation

Use the histogram of gradient directions

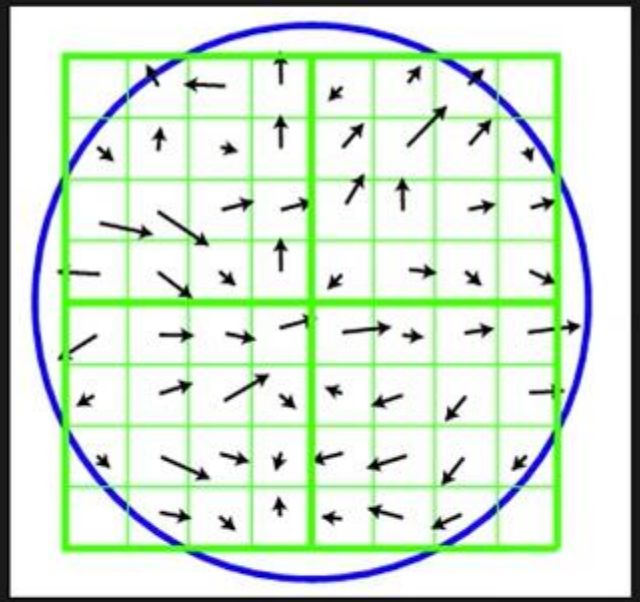


Image gradient directions

$$\theta = \tan^{-1}\left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x}\right)$$

Computing the Principal Orientation

Use the histogram of gradient directions

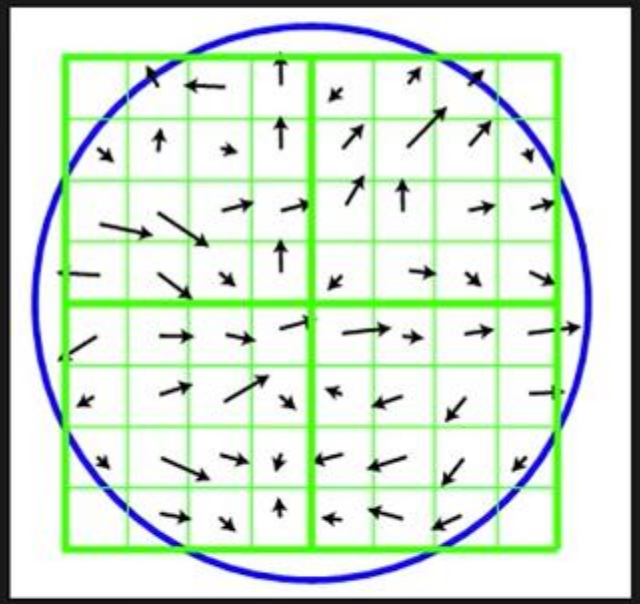
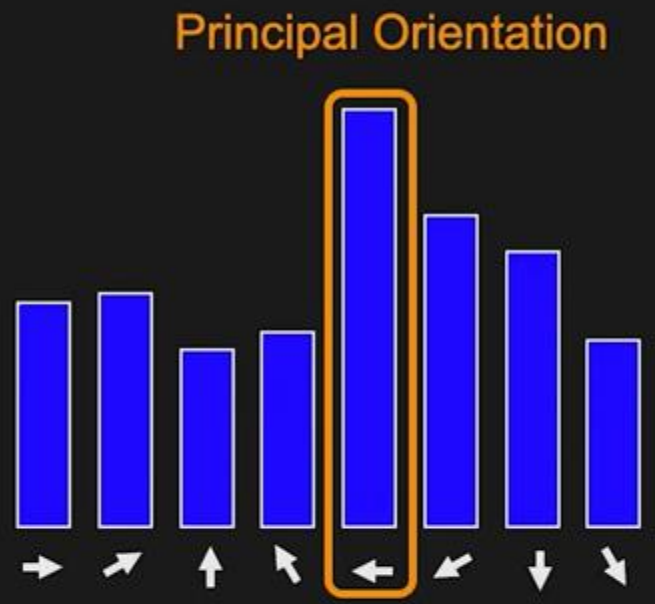


Image gradient directions

$$\theta = \tan^{-1}\left(\frac{\partial I}{\partial y} / \frac{\partial I}{\partial x}\right)$$



Choose the most prominent gradient direction

SIFT Rotation Invariance

Use the principal orientation to undo rotation





SIFT Rotation Invariance

Use the principal orientation to undo rotation



SIFT Descriptor

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Topic: SIFT Detector, Module: Features

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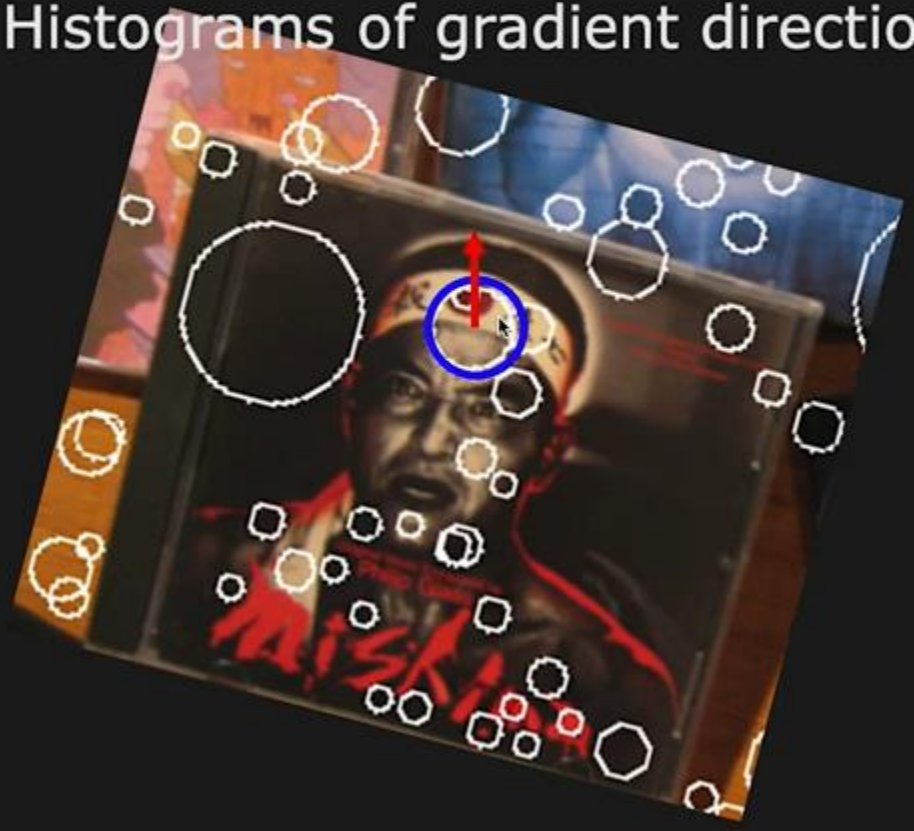
SIFT Descriptor

Histograms of gradient directions over spatial regions



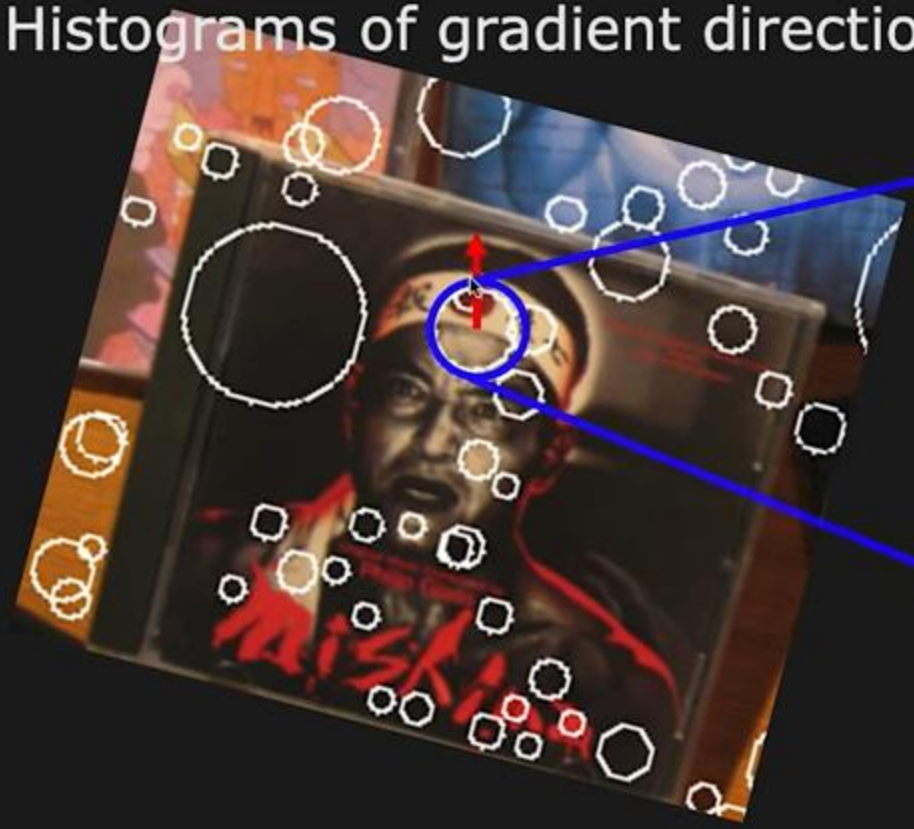
SIFT Descriptor

Histograms of gradient directions over spatial regions



SIFT Descriptor

Histograms of gradient directions over spatial regions



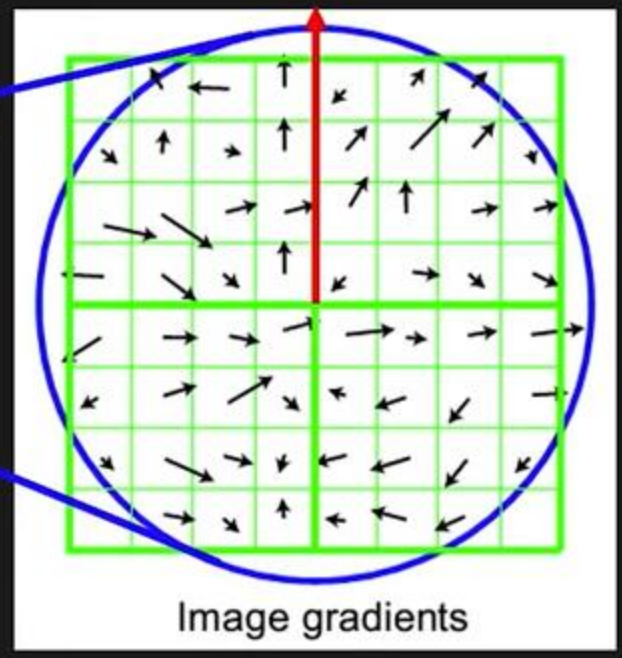
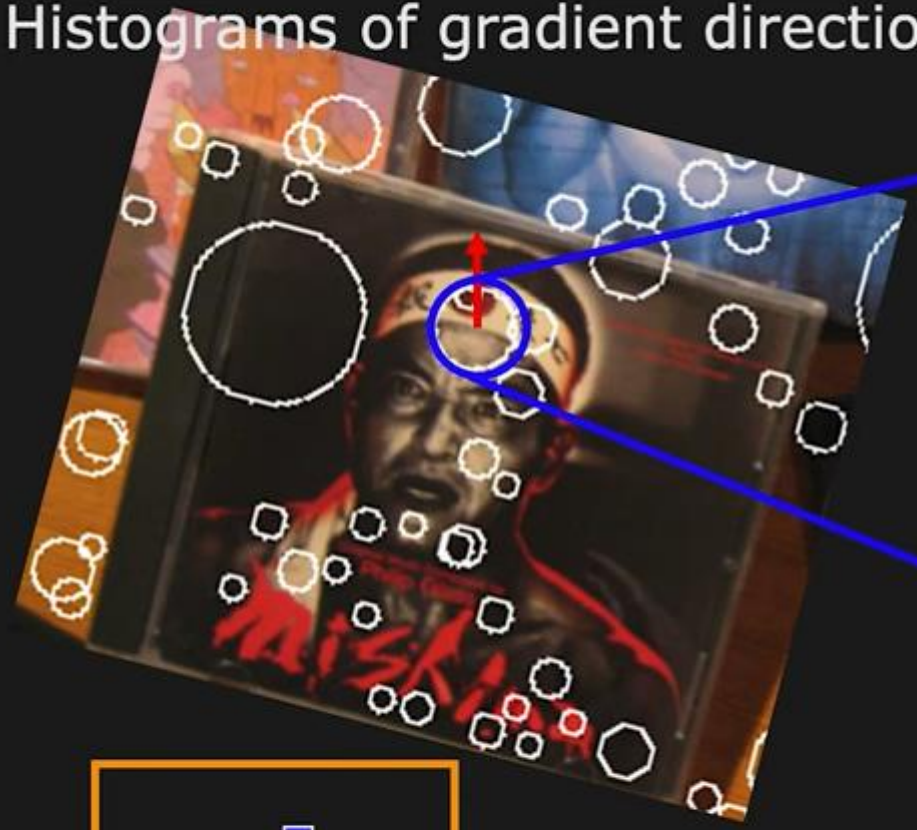


Image gradients

SIFT Descriptor

Histograms of gradient directions over spatial regions



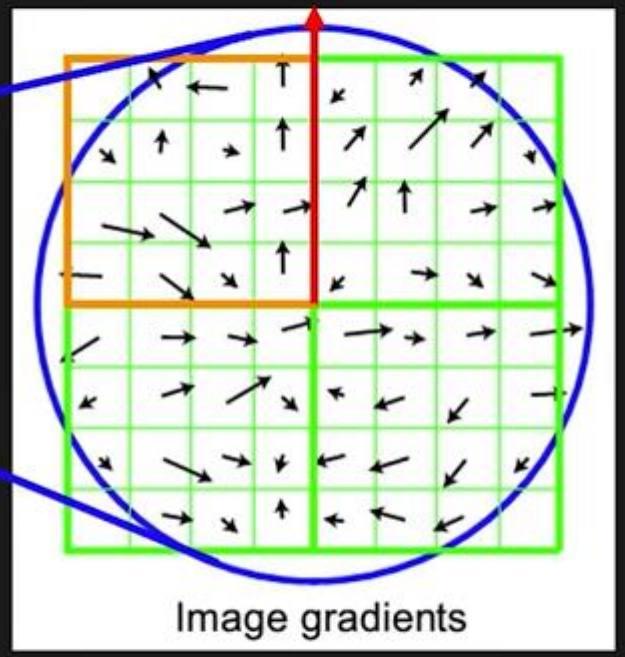
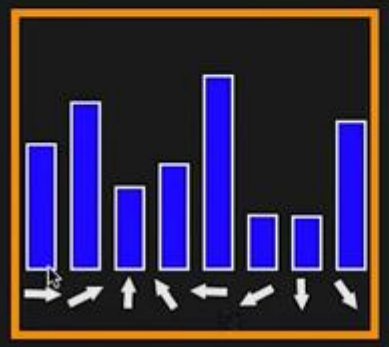


Image gradients



SIFT Descriptor

Histograms of gradient directions over spatial regions

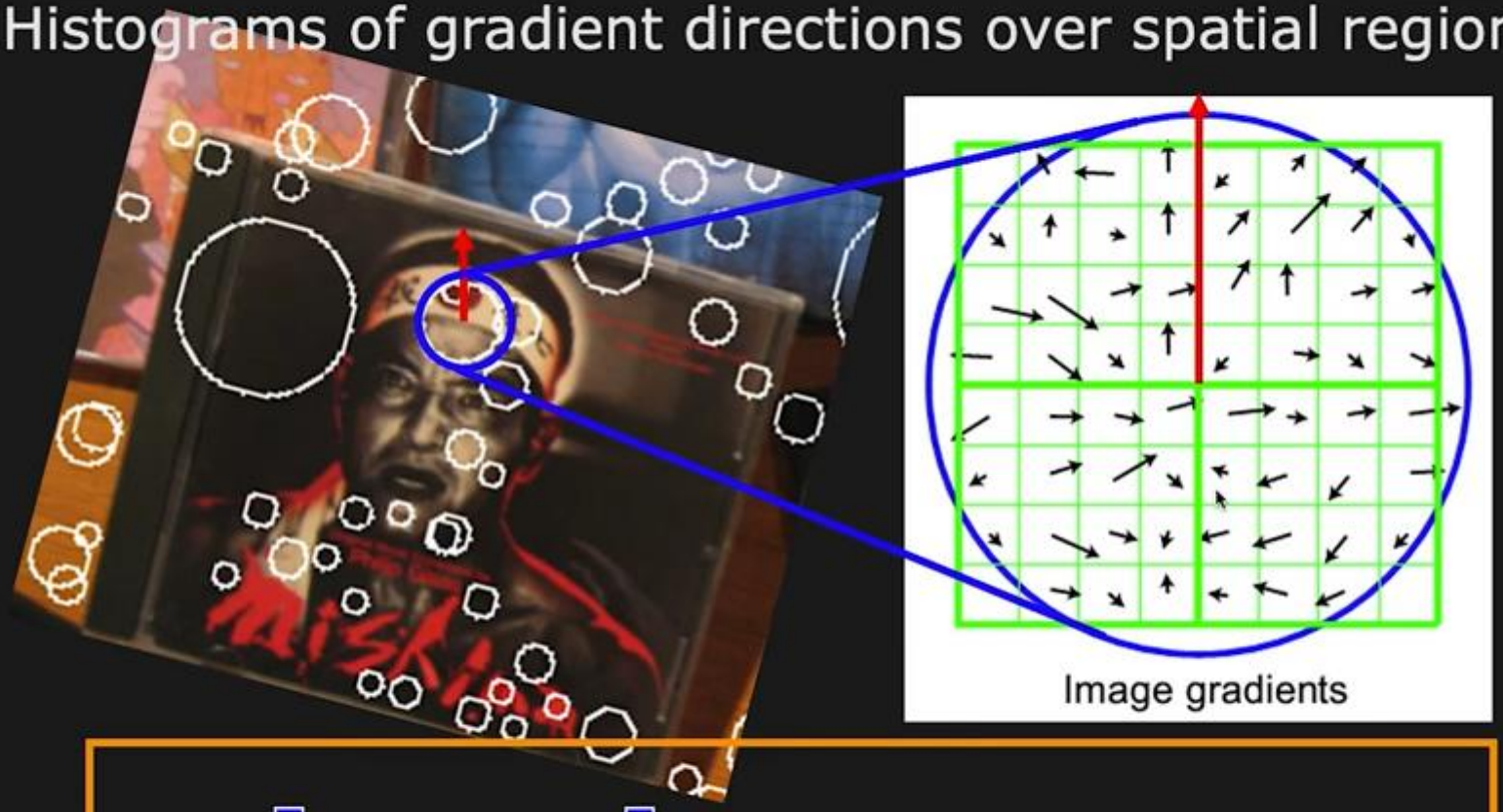
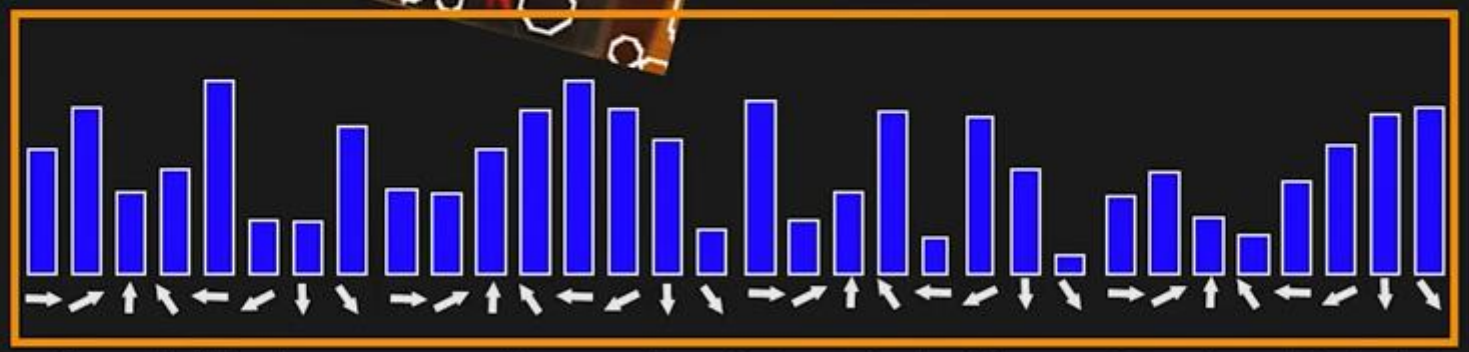


Image gradients



Normalized Histogram: Invariant to Rotation, Scale, Brightness

Comparing SIFT Descriptors

Essentially comparing two arrays of data.

Let $H_1(k)$ and $H_2(k)$ be two arrays of data of length N.

L2 Distance:

$$d(H_1, H_2) = \sqrt{\sum_k \left(H_1(k) - H_2(k)\right)^2}$$

Smaller the distance metric, better the match.

Perfect match when $d(H_1, H_2) = 0$

Comparing SIFT Descriptors

Essentially comparing two arrays of data.

Let $H_1(k)$ and $H_2(k)$ be two arrays of data of length N.

Normalized Correlation:

$$d(H_1, H_2) = \frac{\sum_k [(H_1(k) - \bar{H}_1)(H_2(k) - \bar{H}_2)]}{\sqrt{\sum_k (H_1(k) - \bar{H}_1)^2} \sqrt{\sum_k (H_2(k) - \bar{H}_2)^2}}$$

where: $\bar{H}_i = \frac{1}{N} \sum_{k=1}^{N} H_i(k)$

Larger the distance metric, better the match.

Perfect match when $d(H_1, H_2) = 1$

Comparing SIFT Descriptors

Essentially comparing two arrays of data.

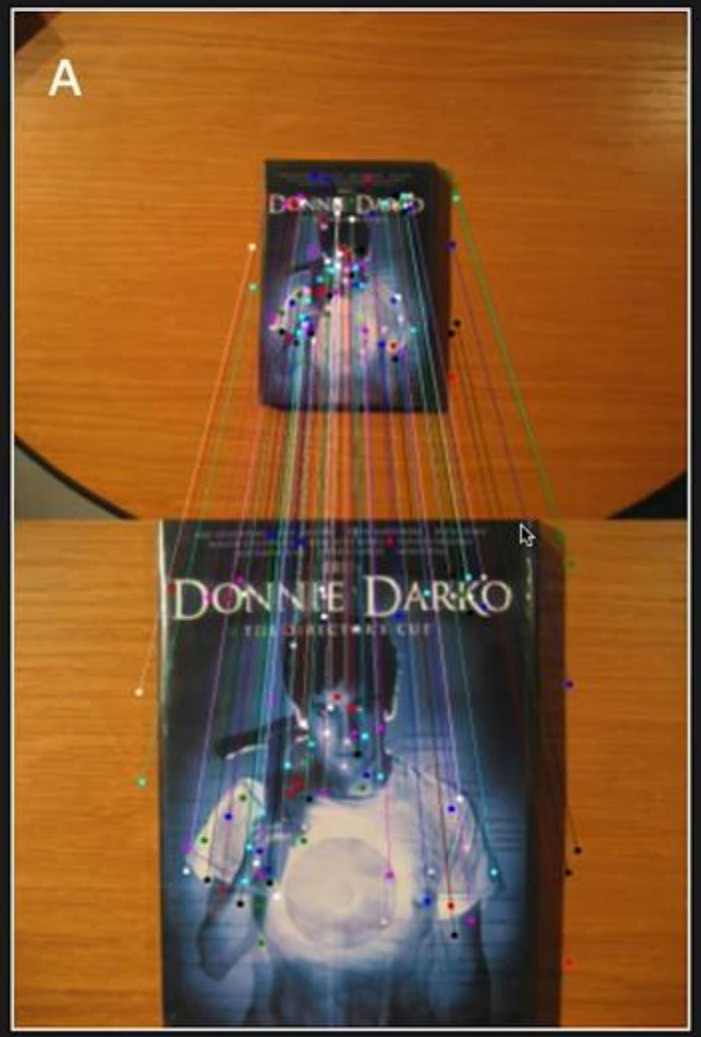
Let $H_1(k)$ and $H_2(k)$ be two arrays of data of length N.

Intersection:

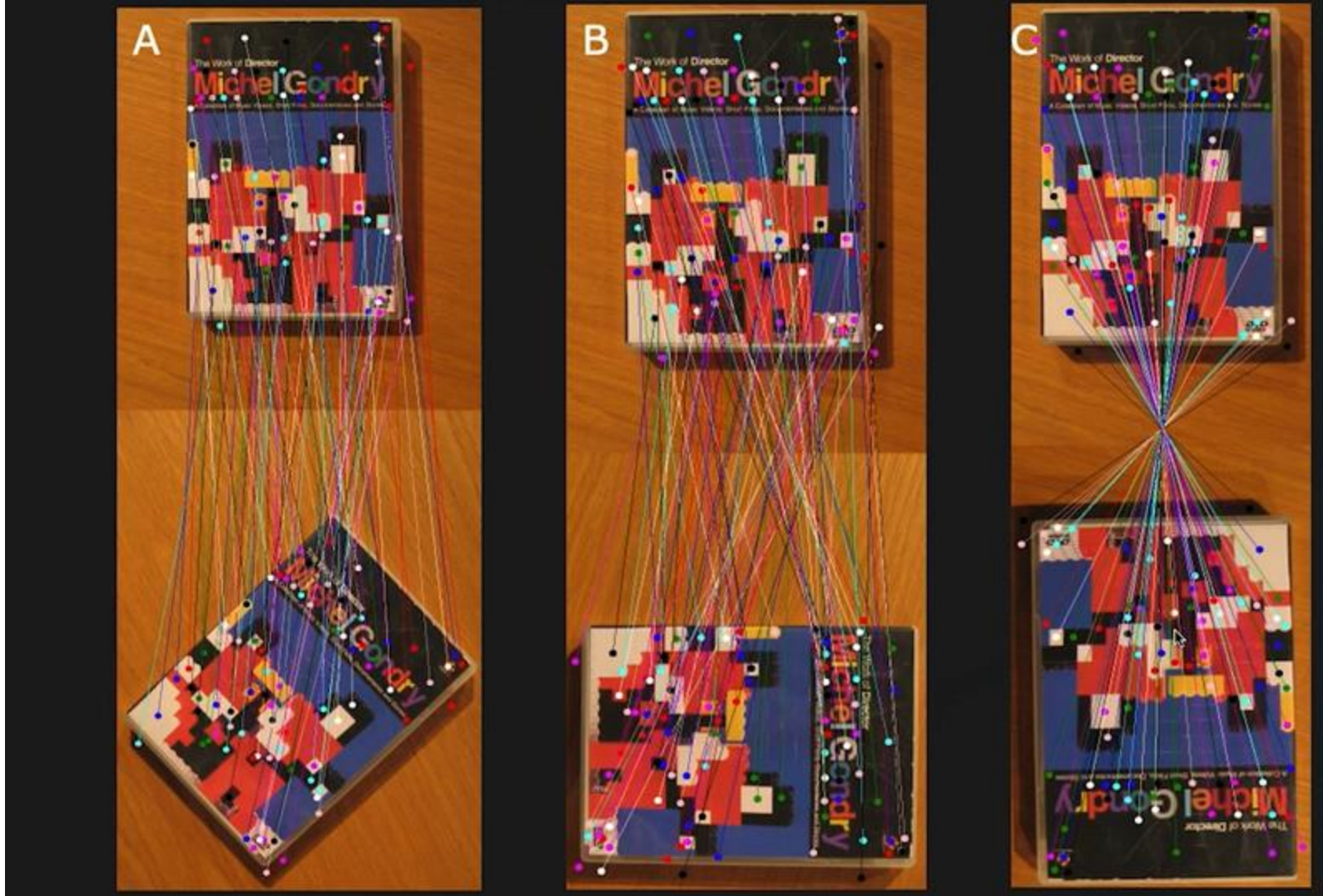
$$d(H_1, H_2) = \sum_k \min\big(H_1(k), H_2(k)\big)$$

Larger the distance metric, better the match.

SIFT Results: Scale Invariance



SIFT Results: Rotation Invariance



SIFT Results: Robustness to Clutter







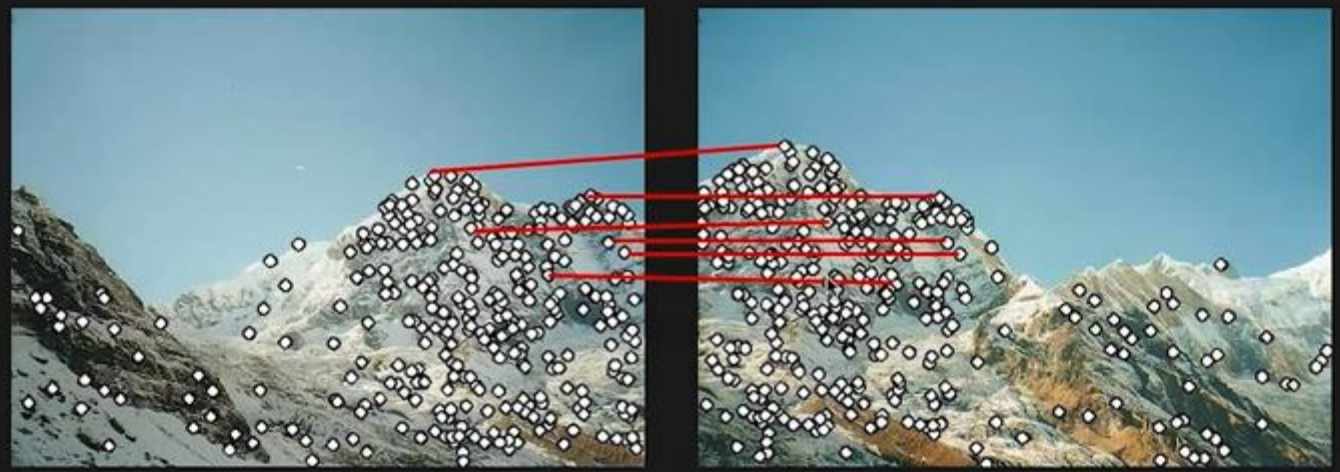
Panorama Stitching using SIFT



Image 1



Image 2



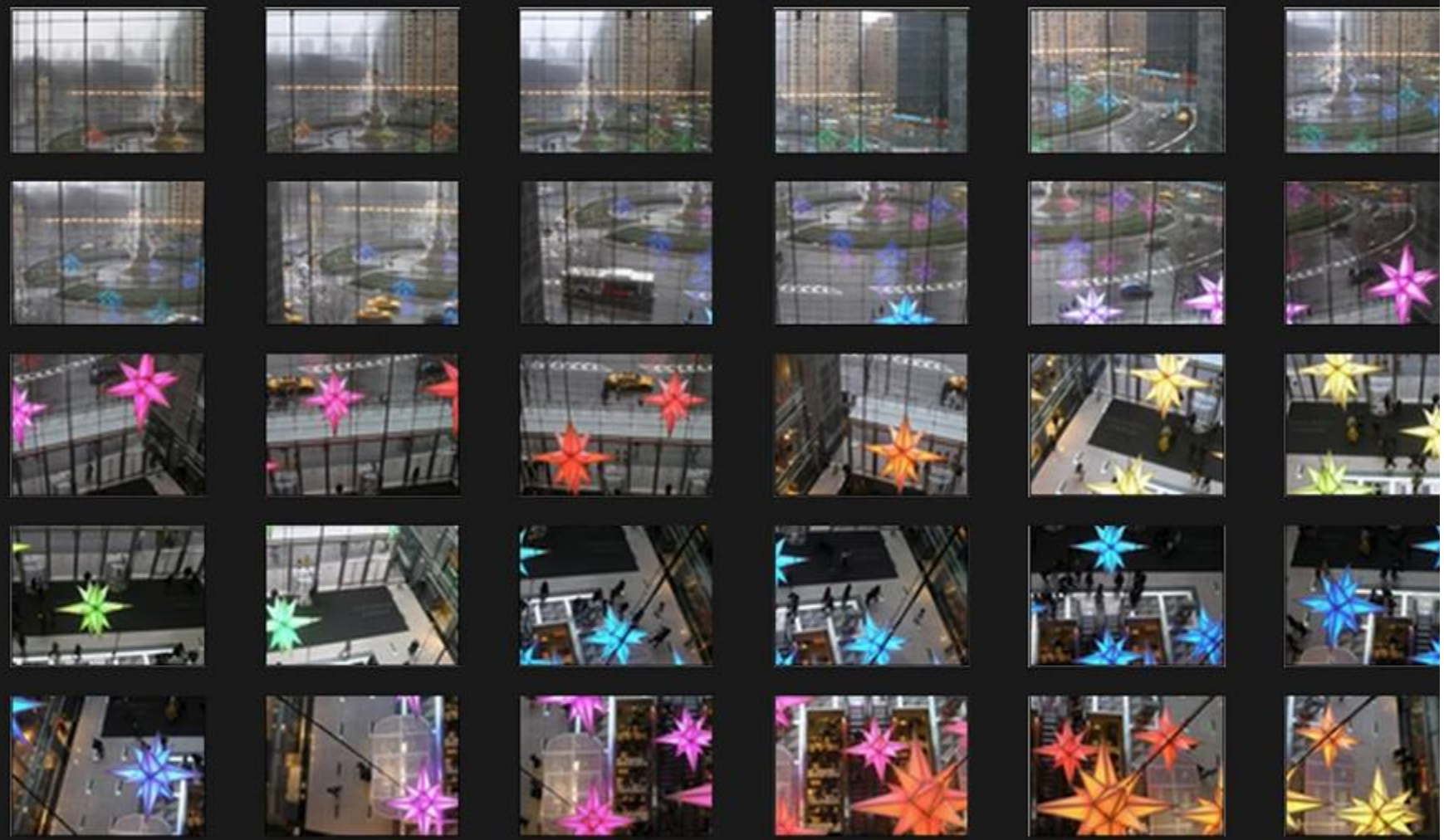
Match SIFT Interest Points

Panorama Stitching using SIFT



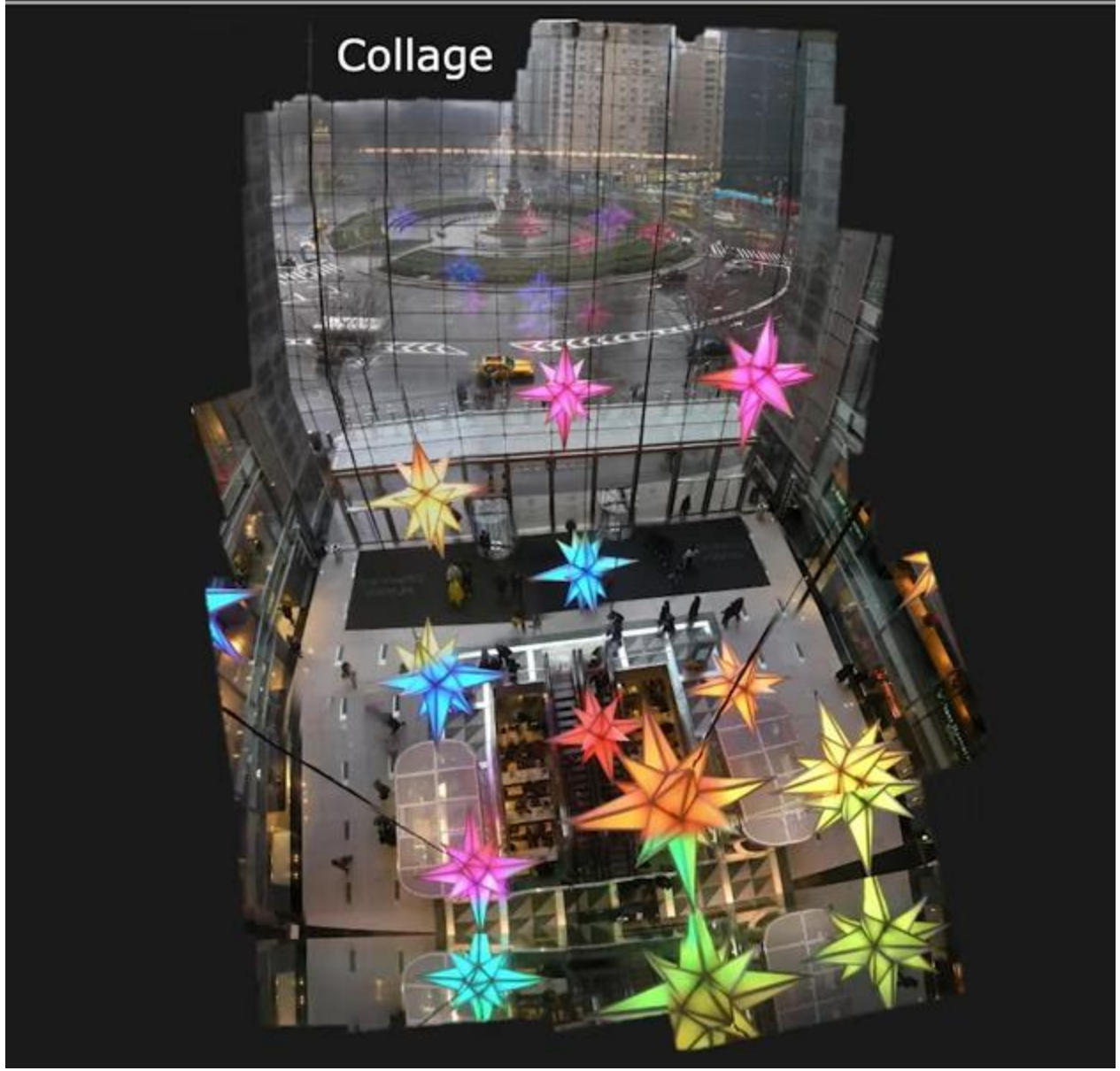
Warp and combine images to create a larger image

Auto Collage using SIFT

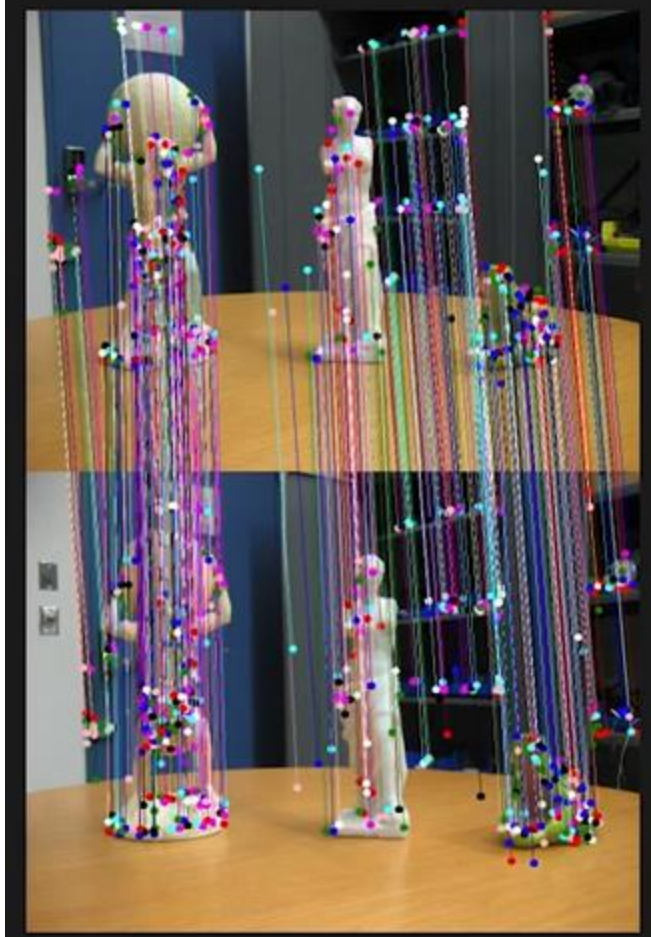


Captured Photos

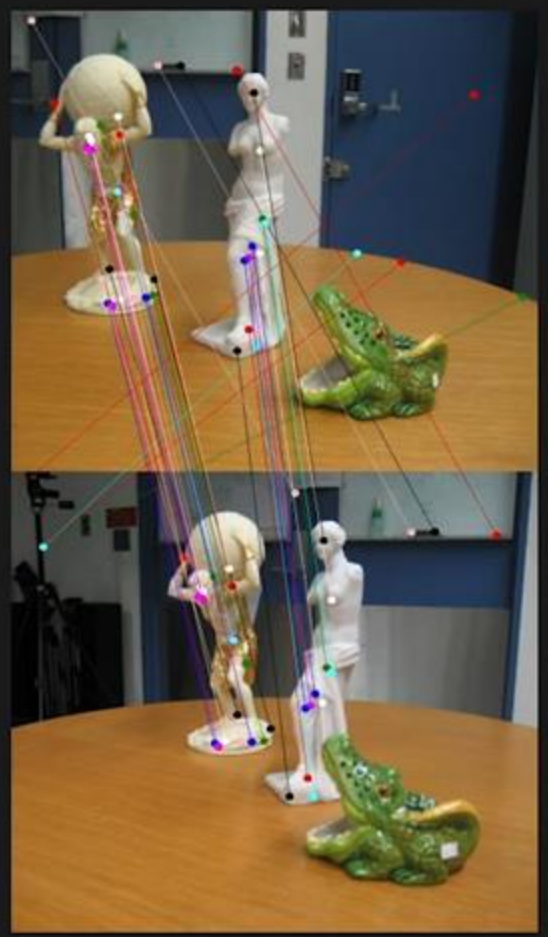
Auto Collage using SIFT



SIFT for 3D Objects?



No Change in Viewpoint



30° Change in Viewpoint



90° Change in Viewpoint

SIFT is reliable for only small changes in viewpoint

Reference

- https://www.youtube.com/watch?v=KgsHoJYJ4S8&list=PL2zRqk16wsdqXEMpHrc4Qnb5rA1Cylrhx&index=12
- https://www.youtube.com/watch?v=wcqbiHonfbo&list=PL2zRqk16wsdqXEMpHrc4Qnb5rA1Cylrhx&index=13
- https://www.youtube.com/watch?v=zItstOggP7M&list=PL2zRqk16wsdqXEMpHrc4Qnb5rA1Cylrhx&index=14
- https://www.youtube.com/watch?v=ram-jbLJjFg&list=PL2zRqk16wsdqXEMpHrc4Qnb5rA1Cylrhx&index=15
- https://www.youtube.com/watch?v=IBcsS8_gPzE&list=PL2zRqk16wsdqXEMpHrc4Qnb5rA1Cylrhx&index=16